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BIOLOGY

Final Revision Guide

CIE IGCSE Biology 0610 | All 20 Chapters + P6 Guide

All 20 Chapters — 2 Pages Each with Full Detail

Key Definitions, Equations & Complete Tables

Exam Tips & High-Yield Facts for Full Marks

Complete P6 Practical Guide + Worked Examples

Index Page with Chapter & Section Page Numbers

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Ch 1 Classification of Living Organisms

1. Characteristics of Living Organisms (MRS GREN)

Letter	Characteristic	Definition
M	Movement	Action by an organism causing change in position or place
R	Respiration	Chemical reactions in cells that break down nutrients to release energy
S	Sensitivity	Ability to detect and respond to changes in the environment (stimuli)
G	Growth	Permanent increase in size and dry mass by cell division or enlargement
R	Reproduction	Process of making more of the same kind of organism
E	Excretion	Removal of toxic materials and waste products of metabolism (e.g. urea, CO_2)
N	Nutrition	Taking in and using materials for energy, growth, and development

2. The Five Kingdoms

Kingdom	Cell Type	Nutrition Mode	Key Feature	Example
Animals	Eukaryote	Heterotrophic (ingest)	Multicellular; no cell walls	Cat, human, insect
Plants	Eukaryote	Autotrophic (photosynthesis)	Multicellular; cellulose walls; chloroplasts	Oak tree, fern, grass
Fungi	Eukaryote	Saprophytic (extracellular digestion)	Chitin cell walls; hyphae structure	Mushroom, yeast, mould
Prokaryotes	Prokaryote	Varied (hetero or auto)	No membrane-bound nucleus; unicellular	Bacteria
Protoctists	Eukaryote	Varied	Mostly unicellular with nucleus	Amoeba, Euglena, Plasmodium

3. Vertebrate Classes

Class	Body Covering	Breathing	Reproduction	Temperature Regulation
Fish	Wet scales	Gills only	Eggs in water (external fertilisation)	Ectotherm
Amphibians	Moist skin	Skin and lungs	Eggs laid in water; external fert.	Ectotherm
Reptiles	Dry scales or scutes	Lungs only	Leathery eggs on land	Ectotherm
Birds	Feathers	Lungs only	Hard-shelled eggs on land	Endotherm
Mammals	Hair or fur	Lungs only	Live birth; young suckled on milk	Endotherm

✎ ***Exam Tip:*** *Viruses are NOT classified in any kingdom. They are non-living: only a protein coat + DNA or RNA. They cannot respire, grow, or carry out metabolic reactions independently.*

Ch 1 Classification - Arthropods, Plants and Keys

4. Invertebrates - Arthropods (all have jointed legs and exoskeleton)

Group	Leg Pairs	Body Segments	Key Distinguishing Features	Examples
Insects	3 pairs	3 (head, thorax, abdomen)	Antennae; most have wings; 3 pairs legs only	Bee, ant, beetle, fly
Arachnids	4 pairs	2 (cephalothorax + abdomen)	No antennae; no wings; 4 pairs legs	Spider, scorpion, tick
Crustaceans	>4 pairs	2 or more	Two pairs of antennae; chalky exoskeleton	Crab, shrimp, woodlouse
Myriapods	Many pairs	Many segments	Long body; 1 or 2 leg pairs per segment	Centipede, millipede

5. Plant Kingdom

Group	How Reproduced	Leaf Venation	Cotyledons	Examples
Ferns	Spores (no seeds or flowers)	Parallel or branched	No seeds	Male fern, bracken
Monocotyledons	Seeds; flowers	Parallel veins	1 cotyledon	Grass, maize, lily, wheat
Dicotyledons	Seeds; flowers	Net-like (branched) veins	2 cotyledons	Rose, oak, bean, sunflower

6. Binomial Naming and Taxonomy Hierarchy

Binomial System: Two-part Latin name: Genus name (capitalised) + species name (lower case). Italicised when typed. e.g. Homo sapiens

Taxonomic Level (broad to narrow)	Example: Human	Example: Domestic cat
Kingdom	Animalia	Animalia
Phylum	Chordata	Chordata
Class	Mammalia	Mammalia
Order	Primates	Carnivora
Family	Hominidae	Felidae
Genus	Homo	Felis
Species	sapiens	catus

7. Dichotomous Keys

Dichotomous key: A tool to identify organisms by working through a series of paired contrasting statements. At each step choose the matching option and follow to next step until a name is reached

- Read BOTH options at each step before choosing
- Base each choice on a single observable, measurable feature
- If using a numbered key: each option directs you to a new numbered pair
- Species have the SAME genus name if they are closely related (e.g. Homo sapiens and Homo heidelbergensis)

✎ ***Exam Tip:*** *In exam questions about classification: if the organism has a nucleus + chitin walls + saprophytic nutrition = FUNGUS. No nucleus at all = PROKARYOTE (bacterium). Unicellular with nucleus = PROTOCTIST.*

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Ch 2 Organisation of the Organism

1. Cell Organelles - Structure and Function

Organelle	Present In	Function
Nucleus	Animal and plant	Contains DNA in chromosomes; controls all cell activities and division
Cytoplasm	Animal and plant	Jelly-like fluid; site of most metabolic reactions; contains dissolved enzymes
Cell membrane	Animal and plant	Partially permeable phospholipid bilayer; controls entry and exit of substances
Mitochondria	Animal and plant	Site of aerobic respiration; produces ATP energy for cell activities
Ribosomes	Animal and plant	Site of protein synthesis; where mRNA is translated into polypeptide chains
Cell wall (cellulose)	Plant only	Fully permeable; provides rigid structural support; prevents cell from bursting by osmosis
Chloroplasts	Plant only (green cells)	Contain chlorophyll; site of photosynthesis; convert light energy to chemical energy
Permanent vacuole	Plant only	Contains cell sap (sugars and mineral salts); maintains turgor pressure; supports plant
Starch grains	Plant cells only	Storage of glucose as insoluble starch for energy reserves

2. Specialised Cells

Cell	Function	Key Adaptations
Ciliated epithelial	Sweep mucus and trapped pathogens upward away from lungs	Cilia beat rhythmically; many mitochondria to power beating
Root hair cell	Absorb water by osmosis and mineral ions by active transport from soil	Very long, thin extension greatly increases surface area
Palisade mesophyll	Main site of photosynthesis in leaf	Tightly packed near top; many chloroplasts; vertical arrangement maximises light exposure
Red blood cell	Transport oxygen bound to haemoglobin	Biconcave disc shape increases surface area:volume; no nucleus = more space for Hb; flexible
Sperm cell	Carry male nucleus to egg for fertilisation	Long flagellum for motility; many mitochondria for energy; acrosome enzyme to penetrate egg
Neuron (nerve cell)	Transmit electrical impulses rapidly	Very long axon; myelin sheath for insulation and speed; synaptic bulbs for communication

✎ ***Exam Tip:*** *Differentiation = process by which unspecialised stem cells develop into specialised cells. Once differentiated, most animal cells cannot change type. Plant cells retain more flexibility (totipotency).*

Ch 2 Organisation - Magnification and Levels

3. Magnification Formula

Magnification (M) = Image Size (I) / Actual Size (A)

What to find	Rearranged formula	Worked example
Magnification M	M = I / A	Image = 50 mm; Actual = 0.05 mm. M = 50/0.05 = x1000
Actual size A	A = I / M	Image = 45 mm; M = x500. A = 45/500 = 0.09 mm
Image size I	I = A x M	Actual = 8 um; M = x200. I = 8 x 200 = 1600 um = 1.6 mm

Unit conversion: 1 mm = 1000 micrometres (um). Always convert to SAME unit before calculating magnification.

4. Levels of Organisation

Level	Definition	Human Body Example
Cell	Basic structural and functional unit of life	Cardiac muscle cell; neuron; red blood cell
Tissue	Group of similar cells working together for one function	Cardiac muscle tissue; squamous epithelium
Organ	Group of different tissues working together for a larger function	Heart (muscle + connective + epithelial tissues)
Organ System	Group of organs cooperating to carry out a major life process	Circulatory system (heart + vessels + blood)
Organism	Complete, independent, living individual	Human being; oak tree; bacterium

5. Stem Cells

Stem cell: Undifferentiated cell that retains the ability to divide and develop into different specialised cell types

Type	Source	Differentiation Potential	Medical Uses and Ethical Notes
Embryonic stem cells	Early embryo (blastocyst, days 4-5)	Pluripotent: can become almost any cell type	Organ/tissue replacement; treat leukaemia. ETHICAL CONCERN: embryo destroyed.
Adult stem cells	Bone marrow; skin; intestinal lining; umbilical cord blood	Multipotent: limited range of related cell types	Bone marrow transplant for leukaemia. FEWER ethical concerns; living donor possible.

6. Reading Scale Bars in Microscope Images

- Step 1: Measure the printed scale bar length in mm using a ruler
- Step 2: Note the real length the scale bar represents (e.g. 10 um or 0.01 mm)
- Step 3: Calculate magnification: M = measured bar (in um) / represented length (in um)
- Step 4: Then use A = I / M to find actual size of any structure in the image

✎ ***Exam Tip:*** *In P6 biological drawings: use sharp HB pencil only. Single continuous lines - no shading. Label lines must be horizontal. Drawing should fill most of the space provided. State the magnification below.*

Ch 3 Movement In and Out of Cells

1. Diffusion

Diffusion: Net movement of particles from a region of HIGHER concentration to LOWER concentration, down the concentration gradient, as a result of random movement. Passive process - no energy required.

Factor	Effect on Rate	Reason
Surface area increases	Rate increases	More surface available for simultaneous diffusion of particles
Temperature increases	Rate increases	Particles have more kinetic energy; move faster; more frequent effective collisions
Concentration gradient increases	Rate increases	Greater difference drives faster net movement in one direction
Diffusion distance increases (thicker)	Rate decreases	Particles must travel further; takes longer to cross the barrier

2. Osmosis

Osmosis: Net movement of WATER MOLECULES ONLY from a region of HIGHER water potential (dilute solution) to LOWER water potential (concentrated solution) through a PARTIALLY PERMEABLE membrane. Passive - no energy required.

Water potential: Tendency of water to move into or out of a solution. Pure water = highest water potential (zero). Adding solutes LOWERS water potential (more negative value).

Solution type	Effect on plant cell	Effect on animal cell
Hypotonic (more dilute than cell - higher WP)	TURGID: water enters by osmosis; cell wall resists and provides support to plant	LYSIS: water enters; no cell wall to resist pressure; cell swells and bursts
Isotonic (same concentration as cell - same WP)	No net water movement; cell stays at normal size and turgor	No net water movement; cell remains normal
Hypertonic (more concentrated than cell - lower WP)	PLASMOLYSIS: water leaves; cytoplasm shrinks away from cell wall; cell becomes flaccid/wilted	CRENATION: water leaves; cell shrivels and wrinkles

3. Active Transport

Active Transport: Movement of particles through a cell membrane from LOWER to HIGHER concentration (against the concentration gradient) using ENERGY from respiration and carrier proteins

Requirement	Detail
Energy (ATP)	Produced by mitochondria during aerobic respiration. Active transport STOPS when respiration is blocked (e.g. no oxygen or cyanide poison)
Carrier proteins	Specific integral proteins in membrane; bind their target molecule and use ATP to transport it against the gradient
Examples	Root hair cells: absorb mineral ions (Mg^{2+}, NO_3^-) against gradient from soil. Villi: absorb glucose even when concentration is higher inside. Kidney tubules: selective reabsorption of glucose from filtrate

✎ ***Exam Tip:*** *Active transport requires MITOCHONDRIA for ATP and CARRIER PROTEINS. Diffusion and osmosis are PASSIVE - no mitochondria needed. This distinction is frequently tested.*

Ch 3 Cell Movement - Comparison and Exchange Surfaces

4. Comparison of Diffusion, Osmosis and Active Transport

Feature	Diffusion	Osmosis	Active Transport
Energy required	No (passive)	No (passive)	Yes - ATP from respiration
Direction	High conc. to low conc.	High WP to low WP	Low conc. to high conc. (against gradient)
Substance moved	Any dissolved particles or gases	Water molecules only	Specific molecules (ions, glucose)
Membrane needed	Not always	Yes - partially permeable	Yes - with specific carrier proteins
Cell examples	O2 into RBC; CO2 from cells	Water into root hair cell	Mineral ions absorbed by roots; glucose in villi
Affected by temperature	Yes (kinetic energy)	Yes (water molecule movement)	Yes (enzyme activity; ATP production)

5. Efficient Exchange Surfaces

All efficient exchange surfaces share these features:

- Large surface area: maximises area available for diffusion
- Thin walls: short diffusion distance (minimises time)
- Good blood supply or ventilation: maintains steep concentration gradient
- Moist surface: gases must dissolve before diffusing through membranes

Exchange Surface	Location in Body	Key Adaptations
Alveoli	Lungs	Millions of tiny sacs = huge SA; walls 1 cell thick; dense capillary network; moist inner surface
Villi + microvilli	Small intestine (ileum)	Finger-like projections with brush border = massive SA; single-cell thick wall; capillaries for glucose and amino acids; lacteals for fats
Root hair cells	Roots	Long thin extension = greatly increased SA; thin cell wall; close to soil water
Leaf mesophyll	Leaves	Spongy air spaces = large SA for CO2 diffusion; thin moist cell walls; stomata for gas exchange
Fish gills	Gill slits	Gill filaments with lamellae; counter-current blood flow maximises O2 uptake

6. Turgor Pressure and Plant Support

Turgor pressure: Pressure exerted by water inside a fully hydrated plant cell pushing outward against the cell wall - provides rigidity and structural support to the plant

- Turgid cells: full of water; high turgor pressure; plant is firm and upright
- Flaccid cells: lost water; no turgor pressure; plant wilts
- Plasmolysed cells: severe water loss; cytoplasm pulls away from cell wall - often irreversible

✎ ***Exam Tip:*** *In P6 osmosis experiments: % change = (final-initial)/initial x 100. Positive % = water gained (hypotonic solution). Negative % = water lost (hypertonic solution). Plot % change (not absolute change) on the y-axis.*

Ch 4 Biological Molecules

1. Chemical Elements

Molecule	Elements Present	Notes
Carbohydrates	C, H, O	H:O ratio approximately 2:1
Lipids (fats and oils)	C, H, O	Less oxygen relative to carbohydrates
Proteins	C, H, O, N, (S)	Nitrogen is essential; sulfur in some (e.g. cysteine)
DNA	C, H, O, N, P	Phosphate in backbone; P unique to nucleic acids

2. Carbohydrates

Name	Monomer	Type	Main Function
Glucose	(itself)	Monosaccharide	Immediate energy source; substrate for respiration
Sucrose	Glucose + Fructose	Disaccharide	Transport in phloem from leaves to other organs
Starch	Glucose	Polysaccharide	Energy storage in plants; insoluble - does not affect osmosis
Glycogen	Glucose	Polysaccharide	Energy storage in animals (liver and muscle cells)
Cellulose	Glucose	Polysaccharide	Structural; forms plant cell walls; provides support and rigidity

3. Lipids (Fats and Oils)

Structure: 1 glycerol molecule + 3 fatty acid chains joined by ester bonds

Function	Detail
Long-term energy storage	More energy per gram than carbohydrates; stored as adipose tissue
Thermal insulation	Subcutaneous fat layer reduces heat loss from body surface
Protection of organs	Fat surrounds and cushions kidneys, heart, and other vital organs
Cell membranes	Phospholipid bilayer forms the basis of all cell and organelle membranes
Fat-soluble vitamins	Vitamins A, D, E, K are dissolved and transported in dietary fats

4. Proteins

Monomer: Amino acids - 20 different types; joined by peptide bonds to form polypeptide chains

Protein Role	Examples	Function in the Body
Enzymes	Amylase, pepsin, lipase, catalase	Biological catalysts; speed up metabolic reactions without being consumed
Structural	Collagen, keratin, actin, myosin	Provide strength, shape, and support to tissues
Transport	Haemoglobin, carrier proteins	Carry molecules (O2, ions, glucose) from one location to another
Antibodies	IgG, IgM, IgA	Produced by B-lymphocytes; bind to specific antigens on pathogens
Hormones (peptide)	Insulin, glucagon, ADH	Chemical messengers coordinating responses between organs

✎ ***Exam Tip:*** *Protein shape is determined by amino acid sequence. If shape changes (denaturation), protein cannot function. This is why enzymes are destroyed by high temperature and extreme pH.*

Ch 4 Biological Molecules - DNA and Food Tests

5. DNA Structure

DNA: Deoxyribonucleic acid - a double-stranded polymer coiled into a double helix, made of nucleotide monomers joined by covalent bonds

Nucleotide: Each unit consists of: phosphate group + deoxyribose sugar + one of four nitrogenous bases (A, T, C or G)

Base Pairing Rules: A pairs with T (2 H-bonds) | C pairs with G (3 H-bonds)

- The two strands run antiparallel and are held together by hydrogen bonds between complementary base pairs
- Memory aid: Apple-Tree (A=T) and Car-Garage (C=G)
- Given strand A-T-C-G-A: complementary strand = T-A-G-C-T
- The base sequence of a gene DIRECTLY determines the sequence of amino acids in the protein it codes for

6. Food Tests - Complete Practical Reference

Substance	Reagent	Positive Result	Negative Result	P6 Key Notes
Starch	Iodine solution	Blue-black	Orange/brown (unchanged)	Add 2-3 drops directly; immediate change. State exact colour.
Reducing sugars	Benedicts + heat in water bath	Brick-red (also orange, yellow, green for less)	Blue (unchanged)	WATER BATH at 80 C for 2 min. NEVER direct flame. Colour intensity shows amount.
Non-reducing sugar (sucrose)	Step 1: boil with HCl. Step 2: cool + add NaHCO3. Step 3: add Benedicts + heat	Brick-red = sucrose was present	Blue = sucrose absent	Must HYDROLYSE sucrose first to get glucose and fructose before testing.
Protein	Biuret: NaOH then CuSO4	Violet/purple	Blue (unchanged)	Add NaOH FIRST then CuSO4. Order MATTERS. Do not reverse.
Lipids	Ethanol emulsion test	Cloudy white emulsion	Clear solution	Shake food with ethanol; pour into cold water; milky cloudiness = lipid present
Vitamin C	DCPIP solution	DCPIP becomes colourless	Purple (unchanged)	Count drops of food liquid needed to decolourise. More drops = LESS vitamin C.

7. Water - The Universal Solvent

Role of Water	Why It Matters
Solvent for reactions	Most metabolic reactions occur in aqueous solution in cytoplasm
Transport medium	Blood plasma (90% water) transports glucose, amino acids, hormones, CO2, urea
Photosynthesis raw material	H2O molecules split to release O2 and protons used to fix CO2
Temperature regulation	Evaporation of sweat from skin removes large amounts of heat energy; stabilises body temp
Hydrolysis reactions	Digestion: water molecules used to break down large molecules (starch, proteins, lipids)

✎ ***Exam Tip:*** *P6 food test rule: ALWAYS state the exact colour you see. Never write positive or negative without the colour. Benedicts brick-red > orange > yellow > green = decreasing sugar concentration.*

Ch 5 Enzymes

1. Properties of Enzymes

Enzyme: A protein that acts as a biological catalyst: speeds up the rate of a chemical reaction without being consumed or permanently changed in the process

Property	Explanation
Made of protein	Chains of amino acids folded into a precise 3D structure; active site is part of this shape
Specific	Active site is complementary to ONE specific substrate only; one enzyme catalyses one reaction
Reusable	Not consumed in the reaction; active site is regenerated and can catalyse the same reaction repeatedly
Temperature-sensitive	Rate increases to optimum then drops to zero (denaturation) at high temperatures
pH-sensitive	Each enzyme has a specific optimum pH; extremes above or below cause denaturation

2. Lock and Key Mechanism

Active site: The specifically shaped region on the enzyme surface where the substrate molecule binds

Enzyme-substrate complex: Temporary structure formed when the substrate fits into the active site

Specificity: The active site shape is COMPLEMENTARY to the substrate - only that specific substrate fits

- Products are released after the reaction; the active site is then free for another substrate molecule
- Denaturation: high temperature or extreme pH breaks bonds holding the enzyme's 3D shape
- The active site changes shape PERMANENTLY - the substrate no longer fits - no reaction

3. Effect of Temperature on Enzyme Activity

Temperature	Effect on Rate	Explanation
Low (0-10 C)	Very slow rate	Low kinetic energy; few effective collisions between enzyme and substrate
Rising to optimum	Rate increases	More kinetic energy; more frequent effective enzyme-substrate collisions
Optimum (~37 C in humans)	Maximum rate	Ideal balance: enough energy + active site shape still intact
Above optimum (>40-45 C)	Rate rapidly falls to zero	Bonds in enzyme break; active site PERMANENTLY changes shape (DENATURATION)

✎ ***Exam Tip:*** *NEVER say an enzyme is killed by heat. Always write: the enzyme was DENATURED - the active site permanently changed shape so the substrate can no longer bind. Denaturation is IRREVERSIBLE.*

Ch 5 Enzymes - pH, Inhibitors, Industrial Uses

4. Effect of pH on Enzyme Activity

Enzyme	Optimum pH	Location	Substrate
Pepsin	pH 2 (acidic)	Stomach (with HCl)	Proteins cut to shorter peptides
Salivary amylase	pH 7 (neutral)	Mouth	Starch hydrolysed to maltose
Pancreatic amylase	pH 7-8 (slightly alkaline)	Duodenum	Starch hydrolysed to maltose
Trypsin	pH 8 (alkaline)	Duodenum	Proteins and peptides to amino acids
Lipase	pH 8	Small intestine	Lipids to fatty acids and glycerol

- pH too high or too low: ionic bonds holding enzyme shape break; active site changes; enzyme denatured; cannot function

5. Effect of Substrate and Enzyme Concentration

Factor	Effect on Rate	Explanation
Increase substrate concentration	Rate increases, then PLATEAUS	All active sites occupied simultaneously (enzyme saturation); no free sites available
Increase enzyme concentration	Rate increases	More active sites available; more E-S complexes forming per second
Reduce substrate (limiting)	Rate falls	Fewer substrate molecules; active sites often unoccupied; less product formed per second

6. Enzyme Inhibitors

Type	How It Works	Can It Be Overcome?	Example
Competitive	Molecule with similar shape to substrate blocks the active site; directly competes	YES: adding more substrate outcompetes inhibitor	Malonate inhibits succinate dehydrogenase
Non-competitive (allosteric)	Binds to a DIFFERENT site on enzyme; distorts active site shape indirectly	NO: not overcome by more substrate; often irreversible	Cyanide; heavy metal ions; nerve agents

7. Industrial Uses of Enzymes

Enzyme	Industry	Specific Application
Amylase	Food and drink	Convert starch to glucose or maltose syrups; bread-making; brewing
Protease	Detergent; food; leather	Biological washing powders (remove protein stains); meat tenderising; cheese production
Lipase	Detergent; food	Remove fat stains; flavour development in cheeses and dairy
Isomerase	Food	Convert glucose to fructose (sweeter taste; used in diet food and drinks)
Pectinase	Fruit juice production	Break down cell wall pectin to increase juice yield and clarity
Lactase (immobilised)	Dairy	Break lactose to glucose + galactose for lactose-intolerant consumers

- Immobilised enzymes: fixed to an inert surface; allow product to flow through; enzyme is reusable; easy separation from product

✎ ***Exam Tip:*** *P6: Rate = 1/time when time is the DV. Graph of rate vs temperature = bell curve (rises to optimum, falls sharply). Graph of rate vs substrate concentration = curve that plateaus at enzyme saturation point.*

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Ch 6 Plant Nutrition

1. Photosynthesis

Photosynthesis: Process by which plants manufacture carbohydrates from inorganic raw materials (CO_2 and H_2O) using light energy absorbed by chlorophyll

Carbon Dioxide + Water --> (Chlorophyll / Light) --> Glucose + Oxygen
$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$

Raw Material	Source	How Obtained by Leaf
Carbon dioxide (CO_2)	Atmosphere (0.04%)	Diffuses through open stomata into intercellular air spaces
Water (H_2O)	Soil water	Absorbed by root hair cells by osmosis; transported up xylem to leaves

Uses of glucose produced:

Use of Glucose	Detail
Respiration	Broken down to release energy (ATP) for all cell activities
Storage as starch	Converted to insoluble starch for storage in leaves, roots, seeds (does not affect osmosis)
Cell wall synthesis	Converted to cellulose for building new cell walls
Transport	Converted to sucrose for long-distance transport in phloem
Protein synthesis	Combined with mineral nitrogen (from nitrate ions) to make amino acids and then proteins
Lipid synthesis	Used to make oils and fats for energy storage in seeds and for membranes

2. Leaf Structure and Adaptations for Photosynthesis

Structure	Function	Adaptation for Photosynthesis
Waxy cuticle	Waterproofing and protection	Transparent: lets light through to palisade cells; reduces water loss by evaporation
Upper epidermis	Transparent protective layer	No chloroplasts; allows light to pass directly to palisade mesophyll
Palisade mesophyll	Main photosynthesis layer	Tightly packed; densely packed with chloroplasts; near top for maximum light interception
Spongy mesophyll	Gas exchange and some photosynthesis	Large air spaces between cells; large SA for CO_2/O_2 diffusion to and from cells
Guard cells and stomata	Regulate gas exchange and water loss	Open in light (turgid); close at night/drought (flaccid); balance photosynthesis vs transpiration
Xylem (in vein)	Water supply to leaf cells	Hollow dead tubes carry water and minerals from roots through the leaf veins
Phloem (in vein)	Export of products	Living sieve tubes carry sucrose away from leaf to rest of plant (translocation)

✎ ***Exam Tip:*** *Broad flat leaves = large surface area for light absorption. Thin = short diffusion distance for CO_2. Transparent upper epidermis = maximum light to palisade. All these features maximise the rate of photosynthesis.*

Ch 6 Plant Nutrition - Limiting Factors and Minerals

3. Limiting Factors of Photosynthesis

Limiting factor: A factor present in such short supply that it restricts the rate of a process even when other conditions are favourable

Limiting Factor	Effect When Increased	Why Rate Eventually Plateaus
Light intensity	Rate increases linearly	Another factor (CO2 or temperature) becomes limiting; rate flattens
CO2 concentration	Rate increases	Light or temperature becomes limiting
Temperature (to optimum)	Rate increases	Above optimum: photosynthetic enzymes (e.g. RuBisCO) denature; rate falls sharply

4. Mineral Ion Requirements

Ion	Essential Function	Deficiency Symptom
Nitrate (NO3-)	Synthesis of amino acids; making chlorophyll indirectly via proteins	Stunted growth; yellowing of older leaves; poor protein production
Magnesium (Mg2+)	Core component of every chlorophyll molecule	Chlorosis: yellowing of young leaves (chlorophyll cannot be made)
Phosphate (PO43-)	Component of DNA, ATP, and cell membranes	Poor root growth; purple/red discolouration of leaves

5. Starch Test on a Leaf (Practical)

Step	Action	Reason
1	Boil leaf in water for about 1 minute	Stops metabolic reactions; breaks down cell membranes
2	Place leaf in boiling ethanol (using water bath, NOT direct flame)	Removes chlorophyll (decolourises leaf) so the iodine colour change will be visible
3	Rinse leaf in cold water	Removes ethanol; rehydrates and softens brittle leaf
4	Spread flat on white tile; add iodine solution	Blue-black = starch present; orange-brown = no starch

6. Measuring the Rate of Photosynthesis (Elodea Method)

Variable	How to Change It	IV / DV / Control
Light intensity	Vary distance of lamp from plant (intensity proportional to 1/distance2)	IV if testing light; DV = bubbles per minute
CO2 concentration	Add different concentrations of NaHCO3 to water	IV if testing CO2; control temperature and light
Temperature	Place beaker in water baths at different temperatures	IV if testing temperature; use same light intensity and CO2 level

• In P6: use distance of lamp as IV (not light intensity directly). Count bubbles per minute as DV. Always keep other variables constant (named specifically).

✎ ***Exam Tip:*** *Rate of photosynthesis plateaus because a DIFFERENT factor becomes limiting - not the same one. When graph flattens for light intensity: CO2 or temperature is now limiting, not light.*

Ch 7 Human Nutrition

1. Balanced Diet - Nutrients and Deficiency Diseases

Nutrient	Main Food Sources	Function	Deficiency Disease/Effect
Carbohydrates	Rice, bread, pasta, potatoes	Primary energy source; fuel for respiration	Fatigue; weight loss if severe
Fats/Lipids	Butter, oils, nuts, oily fish, avocado	Energy store; thermal insulation; cell membranes; absorb fat-soluble vitamins	Poor vitamin absorption if very low fat diet
Proteins	Meat, fish, eggs, dairy, beans, lentils	Growth; repair of damaged tissues; enzymes; antibodies	Kwashiorkor (severe deficiency in children: swollen belly, thin limbs)
Vitamin C	Citrus fruits, peppers, kiwi, broccoli	Collagen synthesis; maintain healthy skin, gums, blood vessels; wound healing	Scurvy: bleeding gums, slow wound healing, joint pain
Vitamin D	Sunlight on skin, oily fish, egg yolk, fortified foods	Absorb calcium from food; mineralise bones and teeth	Rickets in children (soft, bowed leg bones); osteomalacia in adults
Iron	Red meat, liver, spinach, lentils, fortified cereal	Component of haemoglobin in RBCs; oxygen transport	Anaemia: fatigue, breathlessness, pale skin, low RBC count
Calcium	Milk, cheese, yoghurt, fortified plant milk, nuts	Bone and tooth mineralisation; muscle contraction; nerve function	Rickets; weak bones; osteoporosis (fragile bones in adults)
Dietary fibre	Wholegrain bread, vegetables, legumes, fruit	Adds bulk; stimulates peristalsis; prevents constipation	Constipation; increased risk of bowel cancer; diverticular disease

✎ ***Exam Tip:*** *Egestion is NOT excretion. Egestion = undigested food leaving via anus as faeces (food never entered body cells). Excretion = removal of metabolic waste products made BY cells (urea, CO_2). This distinction is frequently tested.*

Ch 7 Human Nutrition - Digestion Enzymes and Absorption

2. Digestive Processes

Term	Definition
Ingestion	Taking food into the body through the mouth
Mechanical digestion	Physical breakdown of food into smaller pieces without chemical change (teeth grinding; bile emulsification of fats)
Chemical digestion	Enzymatic hydrolysis: large insoluble molecules broken down into small soluble molecules
Absorption	Passage of small digested molecules through the intestine wall into the blood or lymph
Assimilation	Digested molecules enter cells and are used in metabolic processes
Egestion	Undigested material expelled from body as faeces via the anus

3. Digestive Enzymes - Where They Work

Enzyme	Substrate	Product	Produced by	Site of Action
Salivary amylase	Starch	Maltose	Salivary glands	Mouth
Pepsin (protease)	Proteins	Shorter peptides	Stomach wall	Stomach (pH 2)
Pancreatic amylase	Starch	Maltose	Pancreas	Duodenum
Trypsin (protease)	Proteins and peptides	Amino acids	Pancreas	Duodenum
Lipase	Fats (lipids)	Fatty acids + Glycerol	Pancreas	Duodenum and small intestine
Maltase, peptidases	Maltose, peptides	Glucose, amino acids	Intestinal wall cells	Small intestine wall

4. Role of HCl and Bile

Substance	Made by / Stored	Role
Hydrochloric acid (HCl)	Stomach wall cells	Kills bacteria in food (pH 2); provides optimum pH for pepsin; denatures proteins uncoiling them for enzyme access
Bile	Liver (stored in gall bladder)	Neutralises stomach acid in duodenum (makes it alkaline for pancreatic enzymes); EMULSIFIES fats into tiny droplets - greatly increases surface area for lipase to act on

5. Absorption in the Ileum (Small Intestine)

Villi: Finger-like projections lining the ileum that massively increase the surface area for absorption

Adaptation	Function
Villi + microvilli (brush border)	Increase total surface area by up to 600-fold; maximise rate of absorption
Epithelium only one cell thick	Very short diffusion distance from gut lumen into blood capillaries
Dense capillary network in each villus	Absorb glucose and amino acids directly into blood; maintain steep concentration gradient
Lacteals (lymph vessels) in each villus	Absorb fatty acids and glycerol which are reassembled into triglycerides and taken into lymph
Good blood supply via hepatic portal vein	Carries absorbed nutrients rapidly to liver for processing; maintains steep gradient in capillaries

6. Large Intestine Function

- Absorbs WATER from remaining indigestible material (mainly cellulose fibre)
- Remaining material = faeces (fibre, dead bacteria, bile pigments, mucus, water)
- Dietary fibre adds bulk and stimulates peristalsis; prevents constipation and bowel disease

✎ ***Exam Tip:*** *Route of fats: fatty acids + glycerol absorbed into lacteals (lymph) as chylomicrons. Route of glucose and amino acids: absorbed into blood capillaries then hepatic portal vein to liver. This distinction matters.*

Clairo

Ch 8 Transport in Plants

1. Xylem and Phloem - Comparison

Feature	Xylem	Phloem
Substance transported	Water and dissolved mineral ions	Sucrose and amino acids (assimilates)
Direction of transport	Upwards only: roots to stem to leaves	Bidirectional: from source to sink
Cell type	Dead cells; no end walls; hollow; lignified walls	Living cells; sieve plates between cells; companion cells provide ATP
Position in stem	Inner side of vascular bundle	Outer side of vascular bundle
Driving process	Transpiration pull (cohesion-tension)	Active loading of sucrose; uses ATP energy
Energy required?	No (passive - driven by evaporation)	Yes (active loading into phloem)

2. Water Uptake and the Transpiration Stream

- Water enters root hair cells by OSMOSIS (soil water has higher water potential than cell contents)
- Root hair cells: long thin extensions greatly increase surface area for absorption
- Water moves: root hair cells -> root cortex cells -> xylem vessels -> up stem -> leaves
- TRANSPIRATION PULL: water evaporates from leaf mesophyll cells -> pulls continuous column of water up xylem via cohesion-tension mechanism

3. Transpiration

Transpiration: Loss of water vapour from a plant: evaporation from the surfaces of mesophyll cells in leaves, followed by diffusion through open stomata to the atmosphere

Factor	Effect on Transpiration Rate	Reason
Temperature increases	Rate increases	Faster evaporation; water molecules have more kinetic energy
Humidity increases	Rate decreases	Lower water potential gradient between leaf air spaces and atmosphere; slower diffusion
Wind speed increases	Rate increases	Removes water vapour from around stomata; steeper WP gradient maintained
Light intensity increases	Rate increases	Stomata open wider in light; more water vapour escapes
Larger leaf surface area	Rate increases	More stomata; more total surface for evaporation

4. Stomatal Control by Guard Cells

Condition	Guard Cell State	Stomata	Effect on Plant
Bright light (daytime)	Turgid (absorb water by osmosis; photosynthesis produces sugars)	Open	CO_2 enters for photosynthesis; O_2 exits; water vapour is lost
Dark (night) or severe drought	Flaccid (lose water; potassium ions pumped out)	Closed	Prevents water loss; limits gas exchange; conserves water

✎ ***Exam Tip:*** *Cohesion = water molecules stick together via hydrogen bonds. Adhesion = water sticks to xylem walls. Together they form a continuous column of water pulled up by transpiration from the leaves.*

Ch 8 Transport in Plants - Translocation and Xerophytes

5. Translocation

Translocation: Movement of sucrose and amino acids in the phloem from regions of production or supply (SOURCE) to regions of use or storage (SINK)

Direction	Source	Sink	When
Downward (and lateral)	Photosynthesising leaf (summer)	Root storage cells; growing fruits	Active growing season
Upward	Root storage (spring mobilisation)	Growing shoot apices and buds	Early spring before leaves out
Any direction	Ripe fruit or leaves (autumn)	Seeds (developing)	Seed filling season

6. Measuring Transpiration - The Potometer

Potometer: Apparatus that measures the rate of water uptake by a cut shoot as an estimate of transpiration rate

Rate = Distance moved by air bubble / Time (mm/min)

Volume rate = Distance x Cross-sectional area of tube (mm3/min)

- Set up entirely under water to prevent air bubbles entering xylem
- Allow equilibration period before starting measurements
- Faster bubble movement = higher transpiration rate
- Test each variable (wind, light, humidity, temperature) while keeping others constant

7. Xerophyte Adaptations (Dry Environment Plants)

Adaptation	How It Reduces Water Loss
Thick waxy cuticle	Waterproof layer on leaf surface reduces direct evaporation through epidermis
Sunken stomata (in pits)	Humid air trapped in pit; reduces water potential gradient between leaf and air
Rolled leaves (e.g. Marram grass)	Stomata face inward; humid air trapped inside roll; protects from wind
Reduced leaf size (or spines in cacti)	Greatly reduces total surface area for transpiration
Deep, extensive root systems	Access groundwater unavailable to shallower-rooted plants
Water-storing tissue (succulents)	Store large water reserves in thick fleshy stems or leaves for dry periods

✎ ***Exam Tip:*** *Potometer measures water UPTAKE not loss directly. Most uptake = transpiration (tiny fraction used in photosynthesis). Faster bubble = more uptake = more transpiration. Equilibrate before each reading.*

Ch 9 Transport in Animals

1. Why Organisms Need a Transport System

- Small organisms (e.g. Amoeba): large SA:Vol ratio allows diffusion to supply all cells
- Large organisms: small SA:Vol ratio; diffusion too slow to reach deep tissues; need circulatory system
- Double circulation: blood passes through heart TWICE per complete circuit of the body

Circuit	Path	Pressure	Purpose
Pulmonary circuit	Right ventricle -> Lungs -> Left atrium	Low pressure	Oxygenate blood; remove CO2
Systemic circuit	Left ventricle -> Body -> Right atrium	High pressure	Deliver O2 and nutrients to all tissues; return CO2 and waste

2. Heart Anatomy and Function

Structure	Location / Feature	Function
Left ventricle	Thickest muscular walls	Pumps blood at HIGH pressure around the entire systemic circuit
Right ventricle	Thinner walls than left	Pumps blood at low pressure to nearby lungs (shorter distance)
Atria (left and right)	Thin-walled upper chambers	Receiving chambers: collect blood returning from circuits; pump into ventricles
Septum	Muscular wall between left and right	Prevents mixing of oxygenated (left) and deoxygenated (right) blood
AV valves (bicuspid and tricuspid)	Between atria and ventricles	Prevent backflow of blood into atria during ventricular contraction
Semilunar valves	At base of aorta and pulmonary artery	Prevent backflow of blood into ventricles after contraction
Coronary arteries	Branch off from aorta at top	Supply cardiac muscle itself with O2 and glucose; blockage = heart attack

3. Blood Vessels - Structure and Function

Vessel	Direction	Wall Structure	Lumen	Special Features
Artery	Away from heart	Thick: muscle layer + elastic tissue + endothelium	Narrow (small)	Elastic recoil maintains pressure between heartbeats; no valves
Vein	Towards heart	Thin: little muscle; elastic tissue; endothelium	Wide (large)	Semi-lunar VALVES prevent backflow; skeletal muscle squeezes blood along
Capillary	Through tissues	One cell thick (endothelium only)	Tiny	Permeable walls allow exchange of substances with tissue fluid

✎ ***Exam Tip:*** *Pulmonary VEIN carries OXYGENATED blood (lungs to heart). Pulmonary ARTERY carries DEOXYGENATED blood (heart to lungs). These are exceptions to the general rule: arteries = oxygenated, veins = deoxygenated.*

Ch 9 Transport in Animals - Blood and Disease

4. Blood Components and Functions

Component	Approx %	Structure	Function
Plasma	~55%	Pale yellow liquid; 90% water	Transports: CO2, urea, hormones, glucose, amino acids, heat, soluble proteins, antibodies, mineral ions
Red blood cells (RBCs)	~44%	Biconcave disc; no nucleus; contains haemoglobin	Transport O2 bound to haemoglobin from lungs to tissues; CO2 partly as carbaminohaemoglobin
Phagocytes (WBCs)	<1%	Large cell; lobed nucleus; granular cytoplasm	Non-specific immunity: engulf and digest pathogens and debris by phagocytosis
Lymphocytes (WBCs)	<1%	Large cell; large round nucleus; little cytoplasm	Specific immunity: B-lymphocytes produce antibodies; T-lymphocytes coordinate immune response
Platelets	<1%	Small fragments of megakaryocyte cells; no nucleus	Blood clotting: convert fibrinogen (soluble) to fibrin (insoluble mesh) at wound site

5. Named Blood Vessels in the Body

Vessel	Contents	Connects
Aorta	Oxygenated blood (highest pressure)	Left ventricle -> body
Pulmonary artery	Deoxygenated blood	Right ventricle -> lungs
Pulmonary vein	Oxygenated blood	Lungs -> left atrium
Vena cava (inferior and superior)	Deoxygenated blood	Body -> right atrium
Hepatic portal vein	Blood rich in absorbed nutrients (variable O2)	Small intestine -> liver (for processing)
Hepatic vein	Blood processed by liver	Liver -> vena cava
Renal artery	Oxygenated blood (high urea)	Aorta -> kidney
Renal vein	Deoxygenated blood (low urea)	Kidney -> vena cava

6. Coronary Heart Disease (CHD)

CHD: Blockage or narrowing of coronary arteries by fatty (cholesterol) deposits (atherosclerosis) -> reduced blood flow to heart muscle -> angina or heart attack

Risk Factor	How It Contributes to CHD
Smoking	Nicotine narrows arteries and raises blood pressure; CO binds haemoglobin reducing O2; damages vessel walls
High saturated fat diet	Raises LDL cholesterol levels; fatty deposits build up inside artery walls
Lack of exercise	Leads to obesity; high blood pressure; weak cardiac muscle; poor circulation
Genetic predisposition	Inherited tendency to high cholesterol or hypertension
High blood pressure	Damages artery wall endothelium; provides site for fatty deposit formation

✎ ***Exam Tip:*** *RBCs are biconcave (not concave on both sides equally) - this gives a larger surface area to volume ratio for faster O2 exchange, and allows them to flex through narrow capillaries without breaking.*

Ch 10 Disease and Immunity

1. Pathogens and Transmission

Pathogen: A microorganism that causes disease

Transmissible disease: A disease in which the pathogen can be passed from one host to another

Pathogen Type	Examples	Diseases	Key Feature
Virus	HIV, influenza, SARS-CoV-2, measles virus	AIDS, flu, COVID-19, measles	Non-living; uses host cell machinery to replicate; no metabolism of its own
Bacterium	Mycobacterium tuberculosis, Vibrio cholerae, Salmonella	TB, cholera, food poisoning	Prokaryote; reproduces independently; some produce toxins
Fungus	Athlete's foot fungus, Candida albicans	Athlete's foot, thrush	Eukaryote; chitin walls; spores spread infection
Protoctist	Plasmodium (carried by Anopheles mosquito)	Malaria	Eukaryote; complex life cycle; requires vector for transmission

2. Modes of Transmission and Prevention

Mode	Disease Example	Prevention Method
Direct contact (blood or body fluids)	HIV/AIDS	Barrier contraception; sterile needles; screen blood supplies
Contaminated food or water (faecal-oral route)	Cholera, Salmonella	Water treatment; sewage treatment; hand washing; cooking food thoroughly
Airborne droplets (respiratory route)	Influenza, TB, COVID-19	Vaccination; face coverings; ventilation; isolation of infected individuals
Vector (animal carries pathogen to host)	Malaria (Anopheles mosquito)	Mosquito nets; insecticide spraying; drain stagnant water; prophylactic drugs

3. Physical and Chemical Defences

Line 1 - Physical (mechanical) barriers:

- Skin: tough, waterproof barrier of keratinised cells; prevents pathogen entry
- Nose hairs and mucus: trap inhaled particles and pathogens
- Cilia in airways: beat rhythmically to sweep mucus (with trapped pathogens) upward and away from lungs

Line 2 - Chemical barriers:

- Stomach HCl (pH 2): kills most bacteria in food and water
- Lysozyme in tears and saliva: enzyme that breaks down bacterial cell walls

✎ ***Exam Tip:*** *Phagocytes are non-specific (attack any pathogen). Lymphocytes are specific (produce antibodies complementary to one specific antigen). Memory cells = faster and stronger response on second exposure.*

Ch 10 Immunity - Antibodies and Vaccination

4. White Blood Cells - Immune Response

White Blood Cell	Mode of Action	Nucleus Shape
Phagocytes	Non-specific response: detect pathogens; engulf them by phagocytosis; digest in lysosomes using enzymes	Lobed (multi-lobed)
B-lymphocytes	Specific: recognise foreign antigens on pathogen surface; produce SPECIFIC complementary antibodies; form memory cells	Large and round
T-lymphocytes	Specific: detect infected cells displaying antigens; coordinate immune response; attack infected cells directly	Large and round

5. Antibodies

Antigen: A protein (or other molecule) on the surface of a pathogen that triggers an immune response in the host

Antibody: A specific Y-shaped protein produced by B-lymphocytes that binds to a complementary antigen

- Agglutination: antibodies cause pathogens to clump together; easier for phagocytes to engulf and destroy
- Neutralisation: antibodies bind to and neutralise toxins or prevent viruses from entering host cells
- Opsonisation: antibodies coat pathogens, flagging them for destruction by phagocytes

6. Types of Immunity

Type	How Acquired	Duration	Memory Cells?	Example
Active natural	Catching and surviving the disease; body produces own antibodies	Long-term (often lifetime)	Yes	Recovering from measles or chickenpox
Active artificial	Vaccination - antigen stimulates immune response without causing disease	Long-term	Yes	MMR vaccine; flu vaccine
Passive natural	Antibodies transferred from mother via placenta or breast milk	Short-term only (weeks to months)	No	Newborn baby protected temporarily
Passive artificial	Injection of ready-made antibodies (antiserum)	Short-term only	No	Antivenom for snakebite; rabies immunoglobulin

7. Vaccination

Vaccine: Contains dead or weakened pathogens, or their antigens - stimulates immune system to produce antibodies and memory cells without causing disease

- Primary response: first exposure -> slow; small antibody production; memory B-cells formed
- Secondary response: second exposure (infection or booster) -> fast; large antibody surge -> disease prevented
- Herd immunity: if enough of population vaccinated -> pathogen cannot spread even to unvaccinated individuals

Type 1 Diabetes: Autoimmune disease: immune system attacks own insulin-producing beta cells in pancreas -> no insulin made -> blood glucose unregulated. Treatment: insulin injections.

✎ ***Exam Tip:*** *Key P6 skill: reading antibody titre graphs. Primary response: slow rise, lower peak. Secondary response: fast rise, much higher peak, slower decline. Passive immunity: no peak delay but fast decline (no memory cells).*

Ch 11 Respiration and Gas Exchange

1. Gas Exchange System Structure

Structure	Function	Key Feature
Nasal cavity	Warm, moisten and filter incoming air	Mucus lining; blood vessels; hair-like cilia
Trachea	Main airway from throat to bronchi	C-shaped cartilage rings prevent collapse; goblet cells secrete mucus; ciliated cells
Bronchi (2)	Branch from trachea; one to each lung	Cartilage rings; ciliated epithelium; smaller diameter than trachea
Bronchioles	Smaller tubes branching within each lung	Smooth muscle controls airflow; no cartilage; very narrow
Alveoli	Site of gas exchange between air and blood	Millions = huge SA; walls 1 cell thick; dense capillaries; moist inner lining

2. Ventilation - Breathing Movements

	Inspiration (breathing IN)	Expiration (breathing OUT)
External intercostal muscles	Contract -> ribs move UP and OUT	Relax -> ribs fall IN and DOWN
Diaphragm	Contracts -> flattens downward	Relaxes -> returns to domed shape
Volume of thorax	INCREASES	DECREASES
Pressure in lungs	DECREASES (below atmospheric)	INCREASES (above atmospheric)
Air movement	Flows IN through airways	Flows OUT through airways

3. Aerobic Respiration

Respiration: Chemical reactions in cells that break down nutrient molecules to release energy as ATP for metabolism

Glucose + Oxygen --> Carbon Dioxide + Water (+Energy as ATP)

$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$

Uses of energy (ATP) released:

- Muscle contraction | Active transport | Protein synthesis | Cell division
- Maintaining body temperature | Growth | Transmission of nerve impulses

4. Anaerobic Respiration

Feature	Yeast (Fermentation)	Human Muscle Cells
Equation	Glucose -> Ethanol + CO_2	Glucose -> Lactic acid
Oxygen required?	No	No
Energy yield	Low (2 ATP)	Low (2 ATP)
Products formed	Ethanol + carbon dioxide	Lactic acid
Practical use	Bread, beer, wine, biofuels	Sprinting; intense exercise when O_2 supply insufficient

*✎ **Exam Tip:** Aerobic respiration: high energy yield. Anaerobic: very low yield. Yeast anaerobic products = ethanol + CO_2. Muscle anaerobic product = lactic acid ONLY (no ethanol or CO_2). Do NOT confuse these.*

Ch 11 Respiration - Oxygen Debt and Smoking Effects

5. Oxygen Debt

Oxygen debt: The extra oxygen needed after vigorous exercise to oxidise the accumulated lactic acid in the liver back to CO_2 and water

- During intense exercise: anaerobic respiration -> lactic acid builds up in muscles
- Lactic acid causes muscle fatigue; aching; reduced contractile force
- After exercise: continued heavy breathing supplies O_2 to liver
- Liver oxidises lactic acid using the extra O_2 -> this is the oxygen debt being repaid
- Heart rate stays elevated after exercise until debt is repaid

6. Alveoli Adaptations for Efficient Gas Exchange

Adaptation	How It Maximises Gas Exchange
Millions of alveoli -> very large total surface area	More surface available simultaneously for O_2 and CO_2 diffusion
Walls only one cell thick (squamous epithelium)	Minimises diffusion distance: O_2 reaches blood almost instantly
Dense capillary network surrounding each alveolus	Blood constantly flowing removes O_2 and delivers CO_2 -> steep gradient maintained at all times
Moist inner surface lining of alveoli	Gases (O_2, CO_2) must dissolve in moisture before diffusing; moist surface enables this
Constant ventilation by breathing	Renews air in alveoli -> maintains high O_2 and low CO_2 inside -> steep gradient always present

7. Effects of Smoking on the Respiratory System

Substance	Effect on Body
Nicotine	Highly addictive; causes vasoconstriction (narrows blood vessels); raises heart rate and blood pressure; damages blood vessel walls
Tar (carcinogens)	Causes DNA mutations in bronchial epithelial cells -> lung cancer; paralyses and destroys cilia -> mucus builds up -> chronic cough; triggers mucus hypersecretion
Carbon monoxide (CO)	Binds to haemoglobin 200-300x more strongly than O_2 -> forms carboxyhaemoglobin -> reduces blood O_2-carrying capacity -> hypoxia; dangerous in pregnancy
Particulates (soot)	Block and irritate airways; trigger asthmatic attacks; contribute to chronic bronchitis

Lung Disease	Cause	Main Effect on Function
Lung cancer	Carcinogens mutate bronchial cell DNA -> uncontrolled proliferation	Tumour blocks airways; reduces gas exchange; often fatal
Emphysema	Tar destroys alveolar walls -> fewer, larger air sacs	Reduced SA for gas exchange -> severe breathlessness on minimal exertion
Chronic bronchitis	Cilia destroyed; goblet cells overactive -> excess mucus	Persistent productive cough; increased infection risk

✎ ***Exam Tip:*** *Inspired vs expired air: expired has more CO_2 (4% vs 0.04%), more water vapour (saturated vs ~0.04%), less O_2 (16% vs 21%), same N_2. Limewater turns milky = CO_2 present. Cobalt chloride paper turns pink = water vapour.*

Ch 12 Coordination and Response

1. Nervous System vs Endocrine System

Feature	Nervous System	Endocrine (Hormonal) System
Type of signal	Electrical impulses along neurons	Chemical hormones in blood
Speed of transmission	Very fast (milliseconds)	Slower (seconds to minutes or hours)
Duration of response	Short-lived; stops when signal stops	Long-lasting; hormone must be broken down
Target specificity	Very precise: specific muscle or gland	More diffuse: many organs respond to same hormone
Reversibility	Immediate reversal when impulse stops	Gradual; requires metabolic removal of hormone

2. The Reflex Arc

Reflex: An involuntary, rapid, automatic response to a stimulus - protects the body without involving conscious thought

Pathway:

Stimulus -> Receptor -> Sensory neuron -> Relay neuron (in spinal cord) -> Motor neuron -> Effector (muscle or gland) -> Response

Component	Type	Function
Receptor	Specialised sensory cell or organ	Detects the stimulus (e.g. pain receptor in skin; rod/cone in eye)
Sensory neuron	Afferent nerve cell	Carries impulse from receptor to CNS (spinal cord or brain)
Relay neuron	Interneuron within CNS	Connects sensory and motor neurons inside the spinal cord
Motor neuron	Efferent nerve cell	Carries impulse from CNS out to the effector organ
Effector	Muscle or gland	Produces the response: muscle contracts or gland secretes

3. The Synapse

Synapse: A tiny junction (gap) between the axon terminal of one neuron and the dendrite or cell body of the next

- Electrical impulse CANNOT jump across the synaptic cleft directly
- Arrival of impulse -> calcium ions enter -> vesicles release NEUROTRANSMITTERS into synaptic cleft
- Neurotransmitters diffuse across and bind to receptors on post-synaptic membrane -> new impulse generated
- Ensures transmission in ONE DIRECTION only (neurotransmitters only on pre-synaptic side)

✎ ***Exam Tip:*** *Reflexes bypass the brain - they use the spinal cord as the relay centre. This makes them FASTER than voluntary responses. Brain becomes aware of reflex AFTER it has occurred. E.g. pulling hand from flame.*

Ch 12 Coordination - Eye, Hormones and Plant Tropisms

4. The Eye - Accommodation and Pupil Reflex

Accommodation (focusing on objects at different distances):

Object Distance	Ciliary Muscles	Suspensory Ligaments	Lens Shape	Effect
Near object	Contract (smaller diameter)	Slacken / go slack	FAT (thick, more convex)	Refracts (bends) light MORE to focus near image on retina
Distant object	Relax (larger diameter)	Taut / pulled tight	THIN (flat, less convex)	Refracts light LESS to focus distant image on retina

Pupil Reflex (controls amount of light entering eye):

Light Level	Circular Muscles	Radial Muscles	Pupil	Purpose
Bright light	Contract	Relax	Constricts (becomes smaller)	Protects retina from damage by excess light
Dim light	Relax	Contract	Dilates (becomes larger)	Allows maximum light in to improve vision

5. Key Hormones

Hormone	Gland	Main Target	Effect
Adrenaline	Adrenal glands (above kidneys)	Heart; liver; muscles; airways	Fight-or-flight: increases heart rate; raises blood glucose; dilates airways; diverts blood to muscles
Insulin	Pancreas (beta cells)	Liver; muscle; adipose cells	Lowers blood glucose: promotes uptake by cells; glycogenesis (glucose -> glycogen storage)
Glucagon	Pancreas (alpha cells)	Liver	Raises blood glucose: glycogenolysis (glycogen -> glucose); gluconeogenesis
ADH	Posterior pituitary gland	Kidney collecting duct	Increases water reabsorption -> produces more concentrated urine; reduces water loss
Testosterone	Testes	Whole body	Male secondary sex characteristics; sperm production; muscle development
Oestrogen	Ovaries	Uterus; whole body	Repairs and thickens uterine lining; female secondary sex characteristics; coordinates menstrual cycle
Progesterone	Ovaries (corpus luteum)	Uterus lining	Maintains thickened uterine lining; prevents menstruation during pregnancy

6. Plant Hormones - Tropisms and Auxin

Auxin: A plant growth hormone that promotes cell elongation - unequal distribution causes bending of roots and shoots

Tropism	Stimulus	Shoot Response	Root Response
Phototropism	Light (from one side)	Bends TOWARDS light (positive): auxin accumulates on shaded side -> more elongation there	Away from light (negative): same auxin distribution but roots more sensitive - auxin inhibits root growth
Gravitropism	Gravity	Bends AWAY from gravity (negative): auxin at lower side promotes more elongation	Bends TOWARDS gravity (positive): auxin at lower side inhibits root elongation -> root grows down

✎ ***Exam Tip:*** *KEY: auxin PROMOTES elongation in shoots but INHIBITS elongation in roots (roots are far more sensitive to auxin). This one difference explains why shoots grow up (towards light, away from gravity) and roots grow down.*

Clairo

Ch 13 Excretion and Homeostasis

1. Excretion vs Egestion

Term	Definition	Example
Excretion	Removal of METABOLIC WASTE - toxic substances made by cells during chemical reactions in the body	CO2 from respiration; urea from amino acid breakdown in liver; salts from sweating
Egestion	Expulsion of UNDIGESTED FOOD as faeces via the anus. This material NEVER entered body cells.	Cellulose fibre; dead bacteria; bile pigments; mucus - all in faeces

2. Excretory Organs

Organ	Substance Excreted	Origin of Waste Product
Lungs	Carbon dioxide (CO2)	Waste product of aerobic respiration in all body cells
Skin (sweat glands)	Water; mineral salts; small amount of urea	Excess water and salts; small metabolic waste in sweat
Kidneys	Urea; excess water; excess mineral salts	Urea: liver deamination of excess amino acids; excess water and salts from diet

3. Urea Formation - Deamination in the Liver

Deamination: The removal of the amino group (-NH2) from excess amino acids in the liver - produces ammonia as an intermediate

- Step 1: Liver cells remove amino group from excess amino acid -> AMMONIA (NH3) + organic acid
- Step 2: Ammonia is VERY toxic; immediately converted to UREA (less toxic) in liver
- Step 3: Urea enters blood; transported to kidneys; excreted in urine

4. Urinary System

Structure	Function
Renal artery	Carries oxygenated blood with HIGH urea concentration INTO the kidney from aorta
Kidney	Filters blood; removes urea, excess water, excess salts; forms urine
Renal vein	Carries blood with LOW urea concentration AWAY from kidney to vena cava
Ureter	Muscular tube carrying urine from kidney down to the bladder
Bladder	Muscular sac that stores urine until urination
Urethra	Tube through which urine exits the body

✎ ***Exam Tip:*** *Urea is made in the LIVER (from amino acid deamination) but excreted by the KIDNEYS. These are different organs. Do NOT say the kidneys make urea - they filter and excrete it.*

Ch 13 Excretion - Nephron and Homeostasis

5. Structure and Function of the Nephron

Stage	Location	Process	Molecules Remaining in Filtrate
Ultrafiltration	Glomerulus -> Bowman's capsule	High blood pressure forces small molecules out of blood into capsule: water, glucose, urea, mineral salts (small molecules pass through)	All small molecules including urea; LARGE molecules (proteins, blood cells) stay in blood
Selective reabsorption	Proximal convoluted tubule; Loop of Henle	Useful substances returned to blood: ALL glucose (active transport); MOST water (osmosis); SOME salts (active transport)	UREA not reabsorbed; remains in filtrate -> concentrated urine

6. Osmoregulation - Controlling Urine Concentration

Condition	ADH Level	Water Reabsorbed	Urine Produced
High water intake (well hydrated)	LOW ADH secreted from pituitary	Less water reabsorbed in collecting duct	Large volume, pale, dilute urine
Dehydration (hot day, sweating, exercise)	HIGH ADH secreted from pituitary	More water reabsorbed in collecting duct	Small volume, dark, concentrated urine
Normal hydration	Medium ADH level	Moderate reabsorption	Normal yellow urine

7. Homeostasis and Negative Feedback

Homeostasis: Maintenance of a constant internal environment (body temperature, blood glucose, water potential) despite changes in external conditions

Negative feedback: When a variable rises above the set point, a response brings it BACK DOWN. When it falls below, a response brings it BACK UP. The response is always OPPOSITE to the change.

Blood Glucose Regulation:

Situation	Hormone Released	Source	Effect on Liver and Cells
Blood glucose rises (e.g. after a meal)	INSULIN	Pancreas beta cells	Stimulates glycogenesis: glucose -> glycogen stored in liver and muscle; increases glucose uptake by cells
Blood glucose falls (e.g. during exercise/fasting)	GLUCAGON	Pancreas alpha cells	Stimulates glycogenolysis: glycogen -> glucose released into blood; gluconeogenesis from amino acids

8. Kidney Failure Treatment

Treatment	How It Works	Advantages	Disadvantages
Kidney dialysis	Machine filters blood through partially permeable membrane; urea diffuses into dialysis fluid; glucose remains	No surgery; no donor needed; immediately available	3x/week sessions; diet restricted; expensive long-term; not a cure
Kidney transplant	Healthy donor kidney surgically implanted; connects to blood supply and ureter	Permanent cure; normal life; no machine dependency	Risk of rejection; immunosuppressant drugs for life; donor shortage; surgery risks

✎ ***Exam Tip:*** *ADH = Anti-Diuretic Hormone (anti = against, diuretic = urine production). More ADH -> less urine (more concentrated). Less ADH -> more urine (more dilute). Negative feedback maintains correct blood water potential.*

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Ch 14 Reproduction in Plants

1. Asexual vs Sexual Reproduction

Feature	Asexual Reproduction	Sexual Reproduction
Number of parents required	1 parent only	2 parents or their gametes
Genetics of offspring	Genetically IDENTICAL to parent (clones)	Genetically DIFFERENT from parents (variation)
Speed	Fast; no mate searching or gamete production	Slower; requires gamete production and fertilisation
Variation produced	None	Yes - due to meiosis and fertilisation; beneficial for evolution
Examples in plants	Potato tubers; strawberry runners; bulbs; cuttings	All flowering plants producing seeds

2. Flower Structure and Function

Part	Male or Female	Function
Anther	Male	Produces pollen grains (containing male gametes); releases pollen when ripe
Filament	Male	Supports and positions anther for pollen release
Stigma	Female	Sticky or feathery platform where pollen grain lands and germinates
Style	Female	Connects stigma to ovary; pollen tube grows down through it
Ovary	Female	Contains one or more ovules; each ovule contains a female gamete (egg cell)
Petals	Neither	Attract insect pollinators through colour, shape, scent
Sepals	Neither	Protect the flower bud before it opens
Nectary	Neither	Produces nectar as a reward that attracts insect pollinators

3. Pollination - Insect vs Wind

Pollination: The transfer of pollen grains from the anther to the stigma of the same or a different plant of the same species

Feature	Insect-Pollinated	Wind-Pollinated
Petals	Large, bright, colourful	Small, dull, often absent or reduced
Scent and nectar	Strong scent; nectar produced	No scent; no nectar
Pollen	Sticky, spiky, or large; produced in small quantities	Smooth, light, dry; produced in very large quantities
Stigma	Sticky surface; inside the flower	Feathery (large SA); hangs outside flower to catch airborne pollen
Anthers	Inside flower (not dangling)	Loosely attached on long filaments; hang outside to release pollen into wind
Examples	Rose, apple, orchid, lavender	Grass, wheat, hazel, corn

✎ ***Exam Tip:*** *Cross-pollination (different plant of same species) increases genetic variation. Self-pollination (same flower or same plant) reduces variation but is more reliable when pollinators are absent.*

Ch 14 Plant Reproduction - Fertilisation, Seeds and Germination

4. Fertilisation and Seed Formation

Fertilisation: The fusion of the male gamete nucleus (from pollen) with the female gamete nucleus (egg cell in ovule) to form a zygote

Step	Event
1	Pollen grain lands on stigma (pollination)
2	Pollen grain absorbs water from stigma; pollen tube begins to grow
3	Pollen tube grows down through style toward ovary
4	Pollen tube enters ovule through micropyle
5	Male gamete nucleus passes down tube; fuses with female egg nucleus -> ZYGOTE (diploid)
6	OVULE develops into SEED | OVARY WALL develops into FRUIT

5. Seed and Fruit Development

Structure Before Fertilisation	Structure After Fertilisation	Contains
Ovule	Seed	Embryo (radicle + plumule) + food store + seed coat (testa)
Ovary wall	Fruit wall (pericarp)	Fleshy, dry, or winged depending on dispersal method
Zygote (inside seed)	Embryo	Radicle = embryonic root; plumule = embryonic shoot; cotyledons = food store

6. Conditions Required for Germination

W - O - W : Water | Oxygen | Warmth

Requirement	Role in Germination
Water	Activates hydrolytic enzymes; hydrolysis of starch and lipid stores -> sugars for respiration; cell expansion
Oxygen	Needed for aerobic respiration to release ATP energy for cell division and growth
Warmth (suitable temperature)	Provides optimum temperature for germination enzymes to function

- LIGHT is NOT required for germination - seeds germinate underground in darkness
- First root to emerge = radicle (grows downward); first shoot = plumule (grows upward)

7. Seed Dispersal Methods

Method	Mechanism	Structural Features	Example
Wind	Carried by air currents	Light; wings; parachute plumes	Dandelion (parachute); Sycamore (propeller wings)
Animal (external)	Hooks catch on fur or clothing	Hooks; barbs; spines	Burdock; goosegrass
Animal (internal)	Fleshy fruit eaten; seeds pass through gut undamaged	Fleshy edible fruit; hard resistant seed coat	Cherry; apple; tomato
Water	Float in water currents or sea	Buoyant fruit; waterproof coat	Coconut
Explosive	Drying causes sudden splitting	Pod dries and twists; seeds flung out	Pea; lupin; Himalayan balsam

✎ ***Exam Tip:*** *Seed dispersal prevents competition between parent plant and offspring for light, water, and mineral ions. Each dispersal mechanism matches the fruit/seed structure - identify the method from the structural features described.*

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Ch 15 Reproduction in Humans

1. Male Reproductive System

Structure	Function
Testes	Produce sperm by meiosis (spermatogenesis); secrete testosterone; located in scrotum (2-3 C cooler than body = optimal for sperm production)
Epididymis	Where newly made sperm mature and are stored; coiled tube attached to each testis
Sperm duct (vas deferens)	Muscular tube that transports sperm from epididymis to urethra during ejaculation
Seminal vesicles and prostate gland	Add fluid to sperm to form SEMEN: provides nutrients (fructose); alkaline pH to neutralise vaginal acid; lubricant
Urethra	Shared passage for urine and semen to exit the body (not simultaneously - sphincter prevents mixing)

2. Female Reproductive System

Structure	Function
Ovaries	Produce eggs (ova) monthly by meiosis (oogenesis); secrete oestrogen and progesterone
Oviduct (Fallopian tube)	Carries egg from ovary to uterus; SITE OF FERTILISATION; cilia line the tube and move egg along
Uterus (womb)	Muscular organ where embryo implants in endometrium (lining) and fetus develops over 9 months
Cervix	Ring of fibromuscular tissue at opening of uterus; dilates during childbirth
Vagina	Receives penis during intercourse; birth canal during delivery

3. Gametes Comparison

Feature	Sperm (Male Gamete)	Egg / Ovum (Female Gamete)
Size	Microscopic; very small	Large; visible to naked eye (~ 0.1 mm)
Mobility	Motile: long flagellum (tail) for swimming	Not motile; moved passively by ciliated oviduct
Number per event	Millions per ejaculation	One released per ovulation (approx. monthly)
Energy reserves	Small; mitochondria in midpiece provide ATP	Large yolk store for nutrition of early embryo before implantation
Chromosome number	Haploid: 23 chromosomes (n)	Haploid: 23 chromosomes (n)

✎ ***Exam Tip:*** *Both sperm and egg are HAPLOID (n = 23). After fertilisation: zygote is DIPLOID (2n = 46). This prevents chromosome number doubling each generation. Meiosis halves chromosome number when gametes form.*

Ch 15 Human Reproduction - Pregnancy, Cycle and Hormones

4. Fertilisation to Implantation

Fertilisation: Fusion of sperm nucleus with egg nucleus in the oviduct -> diploid zygote (46 chromosomes)

Stage	Event	Time
Fertilisation	Sperm meets egg in oviduct; nuclei fuse -> zygote	Day 0
Cleavage	Zygote divides repeatedly by mitosis -> ball of cells (morula -> blastocyst)	Days 1-4
Implantation	Blastocyst burrows into thickened uterine endometrium lining	Days 6-10
Embryo development	Cells differentiate; all organs begin to form; placenta develops from chorionic villi	Weeks 2-8
Fetal development	All organs present and growing; gaining weight; movements felt by mother	Weeks 9-40
Birth	Cervix dilates; uterine contractions; baby and placenta delivered	~Week 40

5. Placenta Functions

Direction	Substances Exchanged	How
Mother -> Fetus	O2; glucose; amino acids; water; mineral ions; antibodies (IgG); vitamins	Diffusion and active transport across placental membrane
Fetus -> Mother	CO2; urea (metabolic wastes)	Diffusion across membrane
Barrier maintained	Mother's and fetus's blood do NOT mix - different blood groups/pressures possible	Separate capillary systems come very close but do not join

Amniotic sac: Membrane enclosing fetus; filled with amniotic fluid which cushions against physical shocks and maintains constant temperature around fetus

6. The Menstrual Cycle (average 28 days)

Days	Event	Hormones Active
1-5	Menstruation: endometrium lining breaks down and is shed (period)	Oestrogen and progesterone are LOW
6-13	Endometrium thickens and repairs; egg follicle matures in ovary	FSH from pituitary stimulates follicle; follicle secretes OESTROGEN (rises)
Day 14	OVULATION: mature egg released from ovary (triggered by LH surge)	LH surge from pituitary triggers egg release
15-28	Corpus luteum (burst follicle) secretes progesterone; lining maintained	PROGESTERONE maintains endometrium; if no fertilisation: progesterone falls
Day 28	If no pregnancy: progesterone falls -> lining breaks down -> new cycle	All hormones fall to low levels

7. Hormones of the Menstrual Cycle

Hormone	Produced by	Role
FSH (follicle stimulating hormone)	Anterior pituitary gland	Stimulates maturation of egg follicle in ovary; stimulates oestrogen production

Hormone	Produced by	Role
Oestrogen	Ovarian follicle	Stimulates endometrium repair and thickening; feeds back to trigger LH surge
LH (luteinising hormone)	Anterior pituitary	Surge triggers ovulation (egg release)
Progesterone	Corpus luteum (post-ovulation)	Maintains endometrium lining; if pregnant: stays high (HCG from embryo maintains corpus luteum)

✎ ***Exam Tip:*** *If fertilisation occurs: embryo produces HCG (human chorionic gonadotropin) -> maintains corpus luteum -> keeps progesterone high -> no menstruation -> pregnancy continues. HCG detected in pregnancy tests.*

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Ch 16 Chromosomes, Genes and Proteins

1. Key Definitions

Term	Definition
Chromosome	A thread-like structure made of DNA and proteins found in the nucleus; carries genes in a linear sequence
Gene	A specific length of DNA that codes for the production of a particular polypeptide (protein)
Allele	One of two or more alternative versions (forms) of a gene that can occupy the same position (locus) on homologous chromosomes
Genome	The complete set of genetic information (all chromosomes and all genes) present in an organism's cells
Diploid (2n)	Cell containing two complete sets of chromosomes (one from each parent). Humans: 2n = 46 (23 pairs)
Haploid (n)	Cell containing one complete set of chromosomes. Human gametes: n = 23

2. DNA Structure

DNA: Deoxyribonucleic acid - a double-stranded polymer in the shape of a double helix, made of nucleotide monomers

Nucleotide: Each consists of: phosphate group + deoxyribose sugar + one nitrogenous base (A, T, C, or G)

Base Pairing Rules: A pairs with T (2 hydrogen bonds) | C pairs with G (3 hydrogen bonds)

- The two DNA strands are held together by hydrogen bonds between complementary base pairs
- Given strand: 3'-A-T-C-G-A-5' Complementary strand: 5'-T-A-G-C-T-3'

3. Protein Synthesis

Stage	Process	Location
Transcription	The DNA double helix unzips; a complementary mRNA strand is synthesised using one DNA strand as template; each base on mRNA is complementary to the DNA base	Nucleus
mRNA exits nucleus	mRNA strand is small enough to pass through nuclear pores into the cytoplasm	Nuclear pores -> cytoplasm
Translation	Ribosome reads mRNA sequence; tRNA molecules bring matching amino acids; amino acids are joined in the correct order to form a specific polypeptide chain	Ribosomes in cytoplasm

mRNA (messenger RNA): Single-stranded RNA copy of one gene; carries the genetic code from nucleus to ribosome

Exam Tip: DNA -> stays in nucleus (too large to leave). mRNA -> carries copy of code to ribosome. Ribosome -> assembles the protein. Amino acid sequence -> directly determined by base sequence of gene. Change in base sequence = change in protein.

Ch 16 Genetics - Inheritance and Cell Division

4. Genetics Terminology

Term	Definition	Example
Dominant allele	Expressed in phenotype when ONE or TWO copies present	B (brown eyes): BB or Bb both give brown eyes
Recessive allele	Only expressed when TWO copies present (homozygous recessive)	b (blue eyes): only bb gives blue eyes
Homozygous	Two IDENTICAL alleles for a gene	BB (homozygous dominant) or bb (homozygous recessive)
Heterozygous	Two DIFFERENT alleles for a gene	Bb (heterozygous; shows dominant trait)
Genotype	The alleles an organism actually has (its genetic makeup)	Bb; AaBb; XY
Phenotype	The observable characteristic produced by the genotype + environment	Brown eyes; normal blood clotting
Carrier	Heterozygous individual who has recessive allele but does not show the condition	Bb - unaffected but can pass b to offspring
Co-dominance	Both alleles are expressed equally in the heterozygote - neither is dominant or recessive	Blood group AB: IAIB - both A and B antigens expressed on RBC surface

5. Cell Division - Mitosis vs Meiosis

Feature	Mitosis	Meiosis
Purpose	Growth; repair; asexual reproduction	Produce gametes for sexual reproduction
Chromosome number	Diploid -> Diploid (2n -> 2n)	Diploid -> Haploid (2n -> n)
Number of divisions	1 division	2 divisions (meiosis I and II)
Cells produced	2 daughter cells	4 daughter cells
Genetic variation	None: daughters genetically identical to parent	Yes: crossing over and independent assortment
Where it occurs	All dividing body (somatic) cells	Gonads only: testes and ovaries

6. Sex Determination

- Female: XX (two X chromosomes). Male: XY (one X and one Y chromosome)
- Eggs (from female): always carry one X chromosome (all eggs = X)
- Sperm (from male): carry either X or Y chromosome (50% X, 50% Y)
- At fertilisation: X sperm + egg (X) = XX (female). Y sperm + egg (X) = XY (male)
- Probability of each sex = 50% in every pregnancy, regardless of previous children

✎ ***Exam Tip:*** *Punnett square: always write gametes in row and column headers. Work out all four combinations. State both genotype ratio AND phenotype ratio. For sex-linked conditions: always write X^Y for males to show only one X allele.*

Ch 17 Variation and Selection

1. Variation

Variation: Differences in the characteristics of individuals of the same species

Cause of Variation	What Causes the Difference	Examples
Genetic only	Different alleles inherited from parents; no environmental influence	ABO blood group; tongue rolling ability; attached/free earlobes; sex
Environmental only	External conditions; no genetic difference	Language spoken; scars from injuries; tattoos; suntan
Both genetic and environmental	Inherited potential modified by environmental conditions	Height (genes set range; nutrition determines final height); skin colour (genes + UV); mass

2. Continuous vs Discontinuous Variation

Feature	Continuous Variation	Discontinuous Variation
Definition	Characters show a range of values with no distinct categories; all intermediate values possible	Characters fall into clearly separate, distinct categories with no intermediates
Genetic control	Many genes acting together (polygenic) + environmental influence	Usually one or two genes; little environmental influence
Graph type	Bell-shaped curve (normal distribution)	Bar chart (separate bars with gaps between categories)
Examples	Height; hand span; mass; IQ; foot size	Blood group (A, B, AB, O); tongue rolling; sex (male/female); flower colour in some plants

3. Mutation

Mutation: A change in the base sequence of DNA - this is the SOURCE of new alleles and new genetic variation in a population

Type	Description	Example
Gene mutation	Change in one or more DNA bases within a gene -> altered amino acid sequence -> different or non-functional protein	Sickle cell anaemia: single base change -> abnormal haemoglobin -> distorted RBCs
Chromosomal mutation	Error in chromosome number during meiosis (non-disjunction) -> gamete with wrong chromosome number	Down's syndrome: trisomy 21 (3 copies of chromosome 21 instead of 2)

Factors that INCREASE mutation rate:

- Ionising radiation: ultraviolet (UV) light; X-rays; gamma radiation; alpha and beta particles
- Chemical mutagens: carcinogens in tobacco tar; some food additives; certain industrial chemicals
- Most mutations are NEUTRAL (silent; no effect). Some are harmful. Rarely: beneficial to organism.

✎ ***Exam Tip:*** *CRITICAL: Mutations happen RANDOMLY and SPONTANEOUSLY - they are not directed by the environment. The antibiotic does NOT cause resistance mutations. The antibiotic SELECTS the pre-existing mutant that happens to be resistant.*

Ch 17 Variation - Adaptive Features and Natural Selection

4. Adaptive Features

Adaptive feature: An inherited characteristic that increases an organism's fitness (ability to survive, reproduce, and pass alleles to offspring) in its specific environment

Plant Type	Environment	Key Adaptive Features
Xerophytes (e.g. Cactus, Marram grass)	Dry/arid/desert - scarce water	Thick waxy cuticle: reduces evaporation | Sunken stomata: trap humid air in pit | Rolled leaves: stomata face inward; reduce wind exposure | Deep roots: access groundwater | Reduced leaf SA or spines: minimise transpiration surface
Hydrophytes (e.g. Water lily, Canadian pondweed)	Aquatic - excess water; submerged or floating	Stomata on upper leaf surface only: gas exchange with air above water | Large air spaces (aerenchyma) in stems: buoyancy and gas storage | Thin or absent waxy cuticle: no need to conserve water | Flexible stems: bend with water current without breaking

5. Natural Selection (Darwin's Theory of Evolution)

Step	Principle	Detail
1	Variation	Random mutations create heritable genetic variation within any population
2	Overproduction	All organisms produce more offspring than can survive given limited resources
3	Competition	Individuals compete for food, water, mates, space, light -> struggle for survival
4	Survival of the fittest	Individuals with the most favourable variations for the current environment survive longer and reproduce more
5	Inheritance	Survivors pass their advantageous alleles to offspring via reproduction
6	Evolution	Over many generations: advantageous alleles increase in frequency; population becomes better adapted

6. Artificial Selection (Selective Breeding)

Artificial selection: Humans choose individuals with commercially or aesthetically desirable traits and breed them together; repeat over many generations

Feature	Natural Selection	Artificial Selection
Who/what does the selecting	Environment: survival pressure	Humans: economic, aesthetic, or practical goals
Speed	Very slow: thousands to millions of years	Fast: significant change in decades to centuries
Genetic variation	Maintained: variation essential for adaptation	Reduced: selective breeding narrows gene pool; inbreeding risk
Goal/benefit	Organism fitness in natural environment	Human economic or practical needs (yield, temperament, disease resistance)
Example	Peppered moth camouflage; antibiotic resistance	High-milk-yield dairy cows; wheat with high grain yield; disease-resistant crops; guide dogs

✎ ***Exam Tip:*** *Antibiotic resistance step-by-step: random mutation -> bacterium gains resistance gene -> antibiotics kill non-resistant bacteria -> resistant bacterium survives -> reproduces -> whole population becomes resistant. Overuse of antibiotics accelerates this process.*

Ch 18 Organisms and their Environment

1. Ecological Terms

Term	Definition
Population	All individuals of ONE species living in the same area at the same time
Community	All populations of DIFFERENT species living together in the same ecosystem at the same time
Ecosystem	A unit containing all the organisms (community) and their non-living (abiotic) environment, interacting together
Habitat	The specific place or environment in which an organism lives (provides food, shelter, breeding sites)
Niche	The ecological role and position of a species in its environment: what it eats, what eats it, when active, where it lives
Producer	An organism that produces organic compounds by photosynthesis (plants, algae) - start of all food chains
Consumer	An organism that obtains energy and nutrients by feeding on other organisms (herbivores, carnivores, omnivores)
Decomposer	Bacteria and fungi that break down dead organic matter and waste - return nutrients to the soil

2. Energy Flow Through Ecosystems

- Sun = principal source of all energy input into biological systems
- Energy flows in ONE direction through food chains - it is LOST at each level, not recycled
- Only about 10% of energy passes to the next trophic level

Energy lost at each level via:

- Heat (generated during respiration) | Movement (muscle contraction) | Excretion / egestion | Uneaten body parts (bones, shells, roots)

Pyramid Type	What It Shows	Shape Always Upright?
Pyramid of numbers	Number of organisms at each trophic level	NOT always - can be inverted (e.g. 1 oak tree feeding thousands of aphids)
Pyramid of biomass	Dry mass of all living organisms at each trophic level	YES - always upright (biomass always decreases up the chain)
Pyramid of energy	Total energy content at each trophic level	YES - always upright (energy always decreases upwards)

✎ ***Exam Tip:*** *Food chain arrows show direction of ENERGY FLOW - not who eats whom. Grass -> Rabbit -> Fox means energy flows FROM grass TO rabbit TO fox. Arrows always point from prey to predator.*

Ch 18 Ecology - Nutrient Cycles and Population Growth

3. The Carbon Cycle

Process	Organisms Involved	Effect on CO2 in Atmosphere
Photosynthesis	Green plants; algae; cyanobacteria	DECREASES CO2: carbon absorbed from air and fixed into glucose
Respiration	All living organisms (plants, animals, fungi, bacteria)	INCREASES CO2: carbon in organic molecules released as CO2
Decomposition	Bacteria and fungi breaking down dead matter	INCREASES CO2: carbon in dead organic matter released
Combustion	Burning fossil fuels; burning wood; forest fires	INCREASES CO2 rapidly: millions of years of stored carbon released
Feeding	Animals consuming plants or other animals	Transfers carbon between organisms in food chain; no net change to atmosphere

4. The Nitrogen Cycle

Nitrogen is essential for: amino acids -> proteins; DNA bases; ATP. Plants CANNOT use N2 gas directly.

Process	Microorganism Involved	Chemical Change
Nitrogen fixation	Rhizobium bacteria (in legume root nodules); free-living Azotobacter	N2 gas (unusable) -> ammonia (NH3) -> nitrates (NO3-) usable by plants
Decomposition (Ammonification)	Decomposer bacteria and fungi in soil	Proteins in dead organisms and waste -> ammonia (NH3)
Nitrification	Nitrifying bacteria in soil	Ammonia -> Nitrites -> Nitrates (NO3-) that plant roots can absorb
Denitrification	Denitrifying bacteria in waterlogged/anaerobic soil	Nitrates -> N2 gas returned to atmosphere (reduces soil fertility)

5. Population Growth - S-Shaped (Sigmoid) Curve

Phase	Description	Birth vs Death Rate
Lag phase	Slow growth: organisms adapting to environment; low reproductive rate	Birth rate slightly > death rate; small population
Log (exponential) phase	Rapid increase: resources abundant; high birth rate; low death rate	Birth rate >> death rate; fastest growth rate
Stationary phase	Population stable: limiting factors acting; birth = death rate	Birth rate = death rate; population at carrying capacity
Death phase	Population declines: resources exhausted; toxic waste builds up	Death rate > birth rate; population falls

Factors that limit population size:

- BIOTIC: food supply; predation; disease; competition for mates; parasitism
- ABIOTIC: temperature; water availability; light intensity; space; mineral ions

✎ ***Exam Tip:*** *Carrying capacity = maximum sustainable population for a given environment. When population reaches carrying capacity, birth rate = death rate and population stabilises. Predator-prey graphs: prey peak PRECEDES predator peak (prey increase first; then predators increase as food is abundant).*

Ch 19 Human Influences on Ecosystems

1. Farming Practices and Environmental Impact

Practice	Environmental Problem	Explanation
Monoculture (single crop over large area)	Reduced biodiversity	Single crop replaces varied habitats; pests thrive; disease spreads rapidly through genetically uniform crop
Artificial fertilisers (nitrates, phosphates)	Eutrophication; water pollution	Excess nutrients leach into waterways during rainfall
Pesticides	Non-target species harmed; bioaccumulation	Toxic to beneficial insects (bees, butterflies); accumulate and concentrate up food chain (biomagnification)
Habitat destruction (land clearance)	Loss of biodiversity; species extinction	Native habitats replaced; organisms lose food, shelter, breeding sites
Factory farming (intensive livestock)	Methane; antibiotic resistance; animal welfare	Cattle and sheep produce methane (potent greenhouse gas); overuse of antibiotics breeds resistant bacteria

2. Eutrophication - Complete Sequence (Must Know in Order)

Step	Event - Learn Each One
Step 1	Nitrates and phosphates from artificial fertilisers LEACH into lake, river, or pond (surface runoff after rain)
Step 2	Algae receive excess nutrients -> grow rapidly -> form dense ALGAL BLOOM on water surface
Step 3	Thick algal bloom BLOCKS SUNLIGHT from reaching aquatic plants below the surface
Step 4	Aquatic plants CANNOT PHOTOSYNTHESISE -> they die
Step 5	AEROBIC BACTERIA decompose large amounts of dead plant material -> bacterial population explodes
Step 6	Bacteria use up all dissolved OXYGEN in the water (BOD - biochemical oxygen demand rises)
Step 7	Fish and all other oxygen-dependent aquatic organisms DIE from lack of dissolved oxygen

3. Air Pollution

Pollutant	Main Source	Environmental and Health Effect
CO2 and methane	Burning fossil fuels; cattle; rice paddies; deforestation	Enhanced greenhouse effect -> global warming -> climate change; sea level rise; extreme weather events
Sulfur dioxide (SO2)	Burning coal and oil (contains sulfur impurities)	Dissolves in atmospheric water -> sulfuric acid rain -> kills trees; acidifies lakes killing aquatic life; erodes limestone buildings
Carbon monoxide (CO)	Incomplete combustion in car engines; gas appliances	Binds irreversibly to haemoglobin (200x stronger than O2) -> reduced O2 transport -> hypoxia; fatal at high concentrations
CFCs (chlorofluorocarbons)	Old refrigerators; aerosol propellants (now mostly banned)	Destroy stratospheric ozone layer -> increased UV radiation reaching Earth -> skin cancer; cataracts; immune suppression
Particulates (PM2.5, PM10)	Diesel engines; industrial chimneys; burning wood	Lung disease; aggravate asthma; contribute to urban smog

✎ ***Exam Tip:*** *Eutrophication: it is the BACTERIA decomposing dead plants that consume oxygen - NOT the algae directly. Fish die because bacteria used up the dissolved oxygen. This two-step mechanism is always tested.*

Ch 19 Human Influences - Deforestation and Conservation

4. Deforestation - Causes and Effects

Effect of Deforestation	Mechanism - How It Happens
Loss of biodiversity	Habitats destroyed; species lose food, shelter, and breeding sites; many go locally or globally extinct
Increased atmospheric CO2	(1) Burning trees releases stored carbon as CO2. (2) Decomposition of felled trees releases CO2. (3) Fewer trees = less CO2 absorbed by photosynthesis. All three contribute.
Disruption of water cycle	Trees return vast amounts of water to atmosphere by transpiration; fewer trees = less evapotranspiration = less rainfall = desertification
Soil erosion and loss of fertility	Tree roots anchor soil; when removed, heavy rain washes away topsoil; dissolved minerals lost (leaching) = infertile land
Increased flooding	Tree roots absorb large volumes of rainwater; without trees, surface runoff increases dramatically -> flash flooding in valleys and plains

5. Conservation Methods

Method	How It Protects Biodiversity
Nature reserves and national parks	Protect entire habitats and all species within from development, hunting, and disturbance
Captive breeding programmes	Breed endangered animals in controlled zoo environments; restore populations; release individuals back to wild
Seed banks (e.g. Svalbard Global Seed Vault)	Preserve seeds of endangered plant species at low temperature; long-term genetic insurance policy for future restoration
Reforestation programmes	Replant trees in deforested areas to restore habitat, stabilise soil, restore water cycle, and sequester carbon
International agreements (e.g. CITES, Paris Agreement)	Ban trade in endangered species; limit carbon emissions; coordinate global conservation effort
Sustainable fishing (quotas, mesh size laws, no-take zones)	Allow fish populations to recover; prevent over-exploitation; maintain marine food webs
Environmental education and ecotourism	Raise public awareness; provide economic value to conservation; fund local conservation through tourism revenue

6. Sustainable Development

Sustainable development: Development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs

- Requires balancing: economic development + environmental protection + social equity
- Examples: renewable energy (solar, wind, hydro); sustainable agriculture; recycling; organic farming; efficient water use

✎ ***Exam Tip:*** *Two reasons why CO2 increases with deforestation: (1) trees burned or decomposed -> CO2 released, AND (2) fewer trees remaining -> less CO2 absorbed by photosynthesis. BOTH reasons needed for full marks.*

Ch 20 Biotechnology and Genetic Modification

1. Biotechnology and Fermentation

Biotechnology: The use of living organisms (or parts of them) to produce substances or carry out industrial and medical processes useful to humans

Organism	Substrate Used	Products Made	Application
Yeast (Saccharomyces cerevisiae)	Glucose or sucrose	Ethanol + CO2 (anaerobic fermentation)	Bread (CO2 causes dough to rise); beer and wine; biofuels
Lactobacillus bacteria	Lactose (from milk)	Lactic acid	Yoghurt; cheese; reduces milk pH below 4.6 causing protein denaturation
Penicillium mould	Growth medium (sugars, minerals)	Penicillin antibiotic	Treating bacterial infections worldwide
GM Escherichia coli	Supplied nutrients in fermenter	Human insulin; HGH; Factor VIII	Medical treatments for diabetes, growth disorders, haemophilia

2. Industrial Fermenters (Bioreactors) - Controlled Conditions

Control Feature	Purpose and How It Maintains Optimum Conditions
Temperature control (water cooling jacket)	Prevents overheating from exothermic reactions; enzymes denature at high temperatures; maintains optimum for maximum yield
pH monitoring and control (buffer addition)	Fermentation produces acids that lower pH; maintains optimum pH for enzyme activity and organism health
Stirrer or paddle system	Ensures even distribution of nutrients, temperature, and oxygen throughout the vessel; prevents settling
Air supply (aerobic) or sealed vessel (anaerobic)	Provides O2 for aerobic fermentation (e.g. yeast for biomass); excluded for anaerobic products (ethanol, lactic acid)
Nutrient inlet (glucose, nitrogen, minerals)	Provides continuous raw materials for growth and product synthesis; maximises yield
Aseptic (sterile) conditions	Prevents contamination by unwanted microorganisms that would compete for nutrients and reduce product yield or quality

✎ ***Exam Tip:*** *Industrial fermenters allow MASS PRODUCTION under precisely controlled conditions. Temperature and pH are monitored continuously and automatically corrected. This ensures maximum enzyme activity and maximum product yield.*

Ch 20 Biotechnology - Genetic Modification

3. Steps of Genetic Modification (GM)

Step	Action	Key Tool or Enzyme
1. Identify	Locate the desired gene in donor organism genome (e.g. human insulin gene on chromosome 11)	Genome sequencing; gene mapping
2. Cut out gene	Cut out the gene from donor DNA using restriction enzymes; leaves short single-stranded 'sticky ends'	Restriction enzymes (molecular scissors) - cut at specific recognition sequences
3. Prepare vector	Cut open a plasmid (small circular DNA from bacteria) using the SAME restriction enzyme -> matching sticky ends produced	Same restriction enzyme as step 2
4. Insert gene	Mix gene + cut plasmid; complementary sticky ends bind by H-bonds; join permanently by covalent bond	Ligase enzyme (molecular glue) - joins sticky ends permanently
5. Transform host	Recombinant plasmid inserted into host bacterium (e.g. E. coli) -> bacterium now contains human gene	Heat shock or electroporation
6. Express and culture	Transformed bacteria grow in fermenter; express inserted gene; produce desired human protein in large quantities	Industrial bioreactor

Restriction enzymes = scissors | Ligase enzyme = glue | Plasmid = vector (carrier)

4. Applications of GM

Application	Gene from	Benefit
Human insulin (Humulin)	Human chromosome 11 (insulin gene)	Identical to natural human insulin; no immune rejection; mass-produced cheaply; treats millions of diabetics
Human Growth Hormone (HGH)	Human HGH gene	Treats growth disorders in children; safer than previous animal-derived HGH (no disease transmission risk)
Herbicide-resistant crops (Roundup Ready)	Bacterial gene for herbicide detoxification	Crops survive herbicide spraying; more effective weed control; higher yields
Bt insect-resistant crops	Bacillus thuringiensis toxin gene	Produces insecticidal protein inside crop plant; kills pest insects; reduces pesticide use
Golden Rice	Daffodil + bacterial genes for beta-carotene	Produces Vitamin A precursor in rice grains; addresses Vitamin A deficiency blindness in developing nations
Drought/flood resistant crops	Genes from stress-tolerant organisms	Enables farming in areas subject to drought or flooding; addresses global food security

5. Advantages and Disadvantages of GM

Advantages of GM	Disadvantages and Concerns
Mass-produce medicines (insulin, vaccines, antibiotics)	GM genes may spread to wild plant relatives via pollen -> uncontrollable 'superweeds'
Crops with higher yield and better nutritional content	Unknown long-term health effects on humans consuming GM food
Crops resistant to drought, flood, pests, and disease	Reduced genetic diversity if GM monocultures dominate agricultural landscapes
Exact human proteins produced -> no immune rejection	Ethical concerns: 'playing God'; animal welfare issues in GM animal production

Advantages of GM	Disadvantages and Concerns
Reduces pesticide and fertiliser use -> less pollution	Patents: biotech companies own GM seeds -> annual licensing fees burden poor farmers in developing countries

✎ ***Exam Tip:*** *Sticky ends are essential: they allow complementary base pairing to join the gene fragment to the plasmid. The SAME restriction enzyme MUST cut both the gene and the plasmid to produce matching complementary sticky ends.*

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P6 Paper 6 - What It Tests and Core Strategy

What Paper 6 Tests

Paper 6 (Alternative to Practical) tests practical skills without requiring actual lab work. You are given experimental descriptions, data, graphs, and results - then answer questions based on these.

Format: 40 marks | 60 minutes | approximately 3 questions (12-15 marks each)

Skill Tested	What You Must Do	Typical Marks
Table construction	Draw ruled table; headings with units in column head only; no units in cells; consistent decimal places	3-4
Graph plotting	Label both axes with quantity AND unit; scale uses more than half grid; even intervals; plot as x or dot; join with ruled lines	4-6
Calculations	Write formula first; show all working clearly; include correct units; round appropriately	1-3
Identifying variables	State IV (what you change), DV (what you measure), controls (NAMED specifically with HOW kept constant)	2-3
Writing conclusions	Link IV to DV; include actual data values from results; do not go beyond the evidence	1-2
Improvements	Specific, named, realistic - e.g. repeat 3 times and calculate mean to improve reliability	1-2
Hazards	Name the hazard AND state the consequence - e.g. HCl is corrosive and could cause skin burns	1
Planning	IV with range; DV with measurement method; named controls; step-by-step method; repeats; safety	4-6
Anomalies	Identify result that does not fit the pattern; circle on graph; do not include in line	1

Core Examiner Priorities

Priority	Rule - Apply to Every Answer
1	Always use DATA: quote specific numbers in every conclusion and evaluation answer - never say results increased without numbers
2	Name control variables explicitly: never say keep everything the same - name EACH variable and HOW it is kept constant
3	Describe not explain: unless the command word is explain, only describe what the data shows
4	Conclusions follow only from evidence: do not state mechanisms or extra biology beyond what the data shows
5	Units every time: in calculations, table headings, and both axis labels on graphs
6	Fully labelled graph axes: both axes need quantity AND unit - missing labels lose marks
7	Improvements must be specific and named: vague answers such as be more careful score zero

Time Management and Key Formulas

- Total: 40 marks in 60 minutes = 1.5 minutes per mark
- Tables and graphs: budget 8-10 minutes each
- Planning questions (6 marks): budget 7-9 minutes

Magnification = Image size / Actual size

Percentage change = (Change / Original value) x 100
Rate = Quantity / Time
Mean = Sum of all values / Number of values
Population estimate = (Number in quadrat / Area of quadrat) x Total area

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P6 Checklists for Tables, Graphs, Planning and Evaluation

Table Checklist

Requirement	Full Detail
Draw all lines with a ruler	Every line in the table - horizontal and vertical - must be drawn with a ruler. Freehand lines lose marks.
Units in column headings only	Format: Temperature / C or Time / s using the slash notation. NEVER write units inside data cells. 10 C in a cell = zero marks for that cell.
Consistent decimal places	All values in the same column must be recorded to the same number of decimal places (e.g. all to 1 d.p.)
IV in first column	Independent variable goes in the leftmost column; dependent variable(s) in subsequent columns to the right
All data correctly placed	Double-check each value is in the correct row and column before moving on

Graph Checklist

Requirement	Full Detail
Both axes fully labelled	BOTH x and y axes need: quantity name AND unit. Example: Concentration / mol dm-3 or Time / s. Missing unit = mark lost.
Scale uses more than half the grid	Points must spread over more than 50% of the grid in both directions. Cramped scale wastes paper and loses marks.
Even scale intervals	Scale MUST increase in equal steps: 2, 4, 6, 8, 10 is correct. 1, 2, 5, 10, 20 is NOT allowed.
Points plotted as small x or dot	Use a small sharp x or neat dot at exact coordinate. Large circles or blobs lose marks.
Lines joined with a ruler	Join consecutive data points with straight ruled lines (unless a smooth curve is specifically justified by the data)
Anomalous points circled and excluded	Circle any anomalous point clearly and draw the line of best fit WITHOUT passing through it

Planning Checklist

Requirement	Full Detail
Independent variable (IV)	Name it AND state the full range and specific values: e.g. temperature of water bath varied from 10 to 50 C in steps of 10 C
Dependent variable (DV)	Name it AND state how it will be measured: instrument + units. e.g. mass of potato strip measured with balance in g
Control variables (at least 2-3)	Name EACH specific variable AND explain HOW it will be kept constant. e.g. same volume of solution 20 cm3 measured with a syringe. NEVER say same conditions.
Method steps	Numbered, logical, step-by-step instructions that any scientist could follow to repeat your experiment
Repeats stated	State: repeat each value 3 times and calculate the mean to improve reliability of results
Safety consideration	Identify a specific hazard AND state the consequence AND precaution taken

Evaluation Checklist

Requirement	Full Detail
Identify specific source of error	Name a precise source of error - NOT just human error. e.g. difficulty judging the exact moment the colour changed

Requirement	Full Detail
Explain the effect on results	State HOW this error affects the results: e.g. this would make timing inconsistent, reducing reliability of the results
Suggest named improvement	Give a specific, practical, named improvement that directly addresses the error: e.g. use a colorimeter to measure colour change objectively
Do not redesign experiment	Improvements should refine the method - not change the fundamental design or swap the entire method

★ ***Mark Scheme:*** *CRITICAL: Hazard answers must have TWO parts: (1) name the hazard AND (2) state what harm it can cause. HCl is corrosive = 0 marks. HCl is corrosive and can cause burns to skin and eyes = 1 mark.*

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P6 Food Tests and Osmosis Investigations

Food Tests - Complete Reference

Substance	Reagent	Positive Result	Negative Result	P6 Key Notes
Starch	Iodine solution	Blue-black	Orange/brown (unchanged)	Add 2-3 drops directly. State blue-black not just colour change. Orange-brown = absent.
Reducing sugars (glucose, maltose)	Benedicts solution + heat	Brick-red (orange / yellow / green if less sugar)	Blue (unchanged)	Heat in WATER BATH at 80 C - NEVER direct flame. Colour intensity shows amount of sugar.
Non-reducing sugar (sucrose)	Step 1: boil with dilute HCl. Step 2: cool and add NaHCO3. Step 3: Benedicts + heat	Brick-red	Blue	MUST hydrolyse sucrose first. HCl breaks sucrose into glucose + fructose. Then test.
Protein	Biuret: NaOH then CuSO4	Violet/purple	Blue	Add NaOH FIRST then CuSO4. Reversing order does not work.
Lipids	Ethanol emulsion test	Cloudy white emulsion	Clear solution	Shake food with ethanol. Pour into cold water. Milky = lipid. Clear = no lipid.
Vitamin C	DCPIP solution (purple-blue)	DCPIP becomes colourless	Stays purple	Count drops to decolourise. More drops = LOWER Vit C concentration.

Interpreting Food Test Results - Rules

- ALWAYS state the exact colour seen - never write positive result or colour change
- For Benedicts: intensity of colour shows AMOUNT of reducing sugar (brick-red = most; green = least)
- A single food sample can contain multiple nutrients - run all relevant tests separately
- State each observation before drawing conclusion: Tube A turned brick-red therefore reducing sugar is present

Osmosis Investigation - Potato or Plant Tissue

Element	Full Detail
Independent variable (IV)	Concentration of salt or sucrose solution (e.g. 0.0, 0.25, 0.5, 0.75, 1.0 mol dm-3)
Dependent variable (DV)	Change in mass or length of potato strip expressed as PERCENTAGE CHANGE (not absolute change)
Control variables	Same initial mass or length; same potato species and tissue type; same temperature; same duration in solution; same surface area of potato; same volume of solution
% Change calculation	% change = (final value - initial value) / initial value x 100. POSITIVE = gained water (hypotonic solution). NEGATIVE = lost water (hypertonic solution).
Graph	Concentration on x-axis vs % change in mass on y-axis. Line crosses x-axis = ISOTONIC POINT (water potential of solution = water potential of cell contents).
Conclusion	Must include the word osmosis AND state direction of water movement AND explain using water potential gradient

★ ***Mark Scheme:*** *Always calculate PERCENTAGE change (not absolute) in osmosis investigations. This allows fair comparison between strips of different initial sizes. NEVER compare raw mass values across different concentrations.*

Enzyme Activity Investigations - Reference

Experiment	DV Measured	Key P6 Notes
Catalase + H2O2 (foam height)	Height of foam produced in mm	Rises from cold to optimum (room temp); drops sharply at high temp (denatured). State denatured not killed.
Lipase + milk + thymolphthalein indicator	Time for blue to colourless in minutes	Rate = 1/time. Higher temp up to optimum = faster. State exact colour: colourless not positive.
Invertase + sucrose then Benedicts	Colour of Benedicts solution	Brick-red = high enzyme activity; more sucrose converted to reducing sugars. Compare colours precisely.
Amylase + starch then iodine at intervals	Time until iodine stays yellow-brown	Test every 30 s with iodine. When no blue-black = starch fully digested.
Yeast + glucose solutions then gas volume	Volume of CO2 gas produced in cm3	Rate = volume/time. Plot TWO labelled lines on same graph for different glucose concentrations.

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P6 Analysis of 6 Real 2025 CIE Papers

Paper 1: 0610/61/M/J/2025 - Diffusion in Agar Cubes

Question	Skill	Key Mark Scheme Points
Q1: Agar cubes (universal indicator) in different HCl concentrations. Time recorded for colour change.	Table; conclusion; improvements; hazards	Table headings: concentration / mol dm-3 and time / s. Units in heading ONLY. Conclusion: As concentration increases time decreases. Improvements: repeat x3 find mean; colorimeter instead of visual judgment; water bath for temperature control.
Q1(b): Plan an investigation into effect of temperature on osmosis in potato strips.	Full planning question	IV: temperature 10-50 C in steps of 10 C. DV: % change in mass. Controls: same conc of solution; same potato species; same initial strip size; same time in solution; same volume of solution.

Paper 2: 0610/62/M/J/2025 - Osmosis with Dilution Calculations

Question	Skill	Key Mark Scheme Points
Q1: Potato discs in 0.0, 0.5, 1.0 mol dm-3 salt solutions. Change in total length measured.	Dilution calculation; % change; line graph; conclusion; evaluation	Dilution: to make 0.5 mol dm-3 in 50 cm3 total: add 25 cm3 of 1.0 stock + 25 cm3 distilled water. % change = (final-initial)/initial x100. Conclusion MUST use word osmosis and state direction of water movement.

Paper 3: 0610/63/M/J/2025 - Catalase and Invertase

Question	Skill	Key Points
Q1: Catalase + H2O2: foam height measured at cold (C), room temp (R), hot (H).	Table from diagram measurements; conclusion; anomaly identification	R = greatest foam height (most active). H = lowest (denatured). State tube letter AND height value. Write: at high temperature the enzyme was denatured - active site changed shape permanently.
Q2: Invertase at 10% and 100% concentration + sucrose. Benedicts test.	Food test interpretation; conclusion	100% -> brick-red. 10% -> green/yellow. Control (water) -> blue (unchanged). State exact colour for each tube. Conclusion: higher invertase concentration produced more reducing sugar.

Paper 4: 0610/61/O/N/2025 - Transpiration in Celery

Question	Skill	Key Points
Q1: Celery stalks in blue dye at 4 C and 40 C. Number of 5 mm sections showing dye staining counted.	Temperature change calculation; table; conclusion; biological drawing of xylem	Temp change = final - initial (include + or - sign). Table headings: water-bath / C and distance dye moved / mm. Conclusion: at higher temperature transpiration rate increased because water molecules have more kinetic energy so evaporate faster. Drawing: sharp pencil; no shading; single continuous lines; horizontal label lines.
Q2: Magnification calculation from microscope image.	M = image size / actual size	Measure image in mm; convert to um (x1000); divide by actual size in um. Write magnification as x500 with no unit.

Papers 5 and 6: O/N/2025 - Respiration and Enzyme Repeats

Paper	Focus	Key Mark Scheme Points
0610/62/O/N/2025	Yeast fermentation: 1% vs 4% glucose concentration; CO2 gas volume measured over time	Rate = volume of gas / time in cm3/min. Plot TWO labelled lines on same set of axes. Conclusion: As glucose concentration increased from 1% to 4% the rate of CO2 production increased from X to Y cm3/min because more substrate was available for yeast enzymes.

Paper	Focus	Key Mark Scheme Points
0610/63/O/N/2025	Invertase investigation (same format as Paper 3)	Reinforces that food tests and enzyme investigations are the HIGHEST FREQUENCY skills. Benedict's colour interpretation and conclusion writing appeared in multiple papers.

Cross-Paper Pattern Analysis - All 6 Papers

Skill or Pattern	Frequency	Priority Level
Table construction with correct headings and units	6/6 papers - ALL	HIGHEST - master first
Conclusion linked to IV and DV using actual data values	6/6 papers - ALL	HIGHEST - master first
Suggesting specific named improvements	6/6 papers - ALL	HIGHEST - master first
Identifying and naming variables (IV, DV, controls)	5/6 papers	HIGH
Percentage change or rate calculation	5/6 papers	HIGH
Osmosis experiments (potato or celery)	4/6 papers	HIGH
Enzyme activity investigations	4/6 papers	HIGH
Line graph plotting	4/6 papers	HIGH
Planning a full investigation	4/6 papers	HIGH
Hazard identification	3/6 papers	MEDIUM
Food tests (Benedicts, iodine, biuret)	3/6 papers	MEDIUM

★ ***Mark Scheme:*** *Master these three skills to secure the most marks: (1) Table construction (2) Conclusion writing using data (3) Evaluation with specific improvements. These appear in EVERY Paper 6 exam.*

P6 Experimental Skills Master Guide

Identifying Variables

Variable Type	Definition	What to Write in Your Exam Answer
Independent variable (IV)	The variable you deliberately change in the investigation	Name the variable AND the full range: e.g. the temperature of the water bath, varied from 10 C to 50 C in steps of 10 C
Dependent variable (DV)	The variable you measure as a result of changing the IV	Name it AND the instrument and units: e.g. the mass of the potato strip, measured with a balance in g
Control variables	All other variables kept constant to make the test fair	Name at LEAST 2-3 specific variables AND explain HOW kept constant: e.g. the same volume of solution 20 cm3 measured with a graduated syringe

★ ***Mark Scheme:*** *NEVER write keep everything the same. Name EACH control variable separately and state HOW it is kept the same. Same conditions = ZERO marks. Same temperature 25 C maintained using a water bath = MARKS.*

Writing Conclusions

Model structure: As [IV] increases/decreases, [DV] increases/decreases. For example when [IV] was [value] the [DV] was [value] compared to [value] when [IV] was [value]. This is because [biological mechanism if asked].

- Always state the DIRECTION of the relationship
- Always quote ACTUAL DATA VALUES - never say results increased without numbers
- Do not go beyond the evidence - only conclude what the data actually shows
- Use correct biological vocabulary (e.g. osmosis; denaturation; photosynthesis)

Reliability, Accuracy and Precision

Term	Definition	How to Improve It
Reliability	Results that can be reproduced when the experiment is repeated under the same conditions	Repeat each value 3 times; calculate mean; identify and discard anomalies before averaging
Accuracy	How close results are to the true or accepted value	Use calibrated instruments; eliminate systematic errors; use appropriate measuring equipment
Precision	How repeatable and consistent results are (small spread)	Use instruments with smaller scale divisions; control variables tightly; careful technique

Evaluation and Improvements - Full Method

Three-step structure: (1) Identify specific error -> (2) Explain its effect -> (3) Name specific improvement

Source of Error	Effect on Results	Named Improvement
Temperature fluctuations in water bath	Enzyme activity varies during experiment making results unreliable and inconsistent	Use an electronic thermostatically-controlled water bath that maintains a constant temperature of X C
Subjective colour judgement (Benedicts end point)	Different observers judge colour change at different moments making timing unreliable	Use a colorimeter set to measure absorbance at a specific wavelength to give objective numerical readings
Variation in size of potato discs	Different surface areas lead to different rates of osmosis making comparisons unfair	Cut all discs to identical dimensions using a cork borer (same diameter) and a ruler (same length)
Single measurement per concentration	One anomalous result cannot be identified or excluded	Repeat each concentration 3 times and calculate the mean; identify and discard anomalous values

Biological Drawing Rules

- Use a SHARP HB pencil only - no pens, biros, or felt tips
- Draw SINGLE CONTINUOUS lines - no sketch marks, no feathering, no gaps mid-line
- ABSOLUTELY NO SHADING of any kind unless stippling is explicitly permitted
- Labels drawn with HORIZONTAL lines (never diagonal) ending clearly at the structure
- Drawing should fill most of the available space - not tiny in one corner
- State magnification AND title below the drawing

✎ ***Exam Tip:*** *Hazard answer template: [Substance/action] is [hazard classification] and could cause [specific harm]. e.g. Hydrochloric acid is corrosive and could cause burns to skin and eyes if it makes contact.*

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P6 Chapter-by-Chapter P6 Practical Skills

Ch	Topic	P6 Practical Skills and Exam Technique Points
1	Classification	Rarely tested as direct practical. May appear as observation of organism characteristics. Use MRS GREN terminology to describe signs of life.
2	Cell Organisation	Microscopy and biological drawing. Magnification = image/actual. Use sharp pencil; no shading; single lines; horizontal labels. Convert 1 mm = 1000 um before calculating.
3	Cell Movement	HEAVILY TESTED. Osmosis in potato or celery. Always use % change not absolute change. IV = concentration. DV = % change in mass or length. Name at least 3 controls with HOW.
4	Biological Molecules	CORE food tests in every exam. Always state EXACT colour. Starch+iodine = blue-black. Reducing sugar+Benedicts+heat = brick-red. Protein+Biuret = violet. Lipid+ethanol = cloudy emulsion.
5	Enzymes	HEAVILY TESTED. Catalase, lipase, invertase, amylase investigations. NEVER write killed - always DENATURED. Rate = 1/time. Bell curve for temperature. Plateau for substrate concentration.
6	Plant Nutrition	Elodea bubble counting for light intensity. Starch test on leaves. Use distance of lamp as IV (not intensity). Describe exact colours in starch test: green region turned blue-black; white region remained orange-brown.
7	Human Nutrition	Food test practicals and interpretation. State result for EACH tube before conclusion. Never jump straight to conclusion - describe observation first then conclude.
8	Plant Transport	Transpiration potometer. Rate = bubble distance / time in mm/min. Drawing xylem cross-section in celery. IV = wind speed / humidity / temperature / light intensity. DV = rate of bubble movement.
9	Animal Transport	Heart rate investigation. Line graph: time on x-axis; heart rate in bpm on y-axis. Describe trends with actual values. Explain why heart rate increases during exercise (muscles need more O2 and glucose).
10	Immunity	Antibody titre graphs. Describe lag phase, peak, and rate of decline with actual values from graph. Compare primary response (slow, low peak) vs secondary response (fast, high peak).
11	Respiration	FREQUENTLY TESTED. Yeast respiration with bubble counting or gas volume measurement. Limewater turns milky = CO2. Aerobic equation from memory. Anaerobic in yeast: glucose -> ethanol + CO2.
12	Coordination	Ruler drop reaction time test. Table with minimum 3 repeats + mean. Identify anomaly. Graph of response vs stimulus intensity. Calculate mean correctly.
13	Homeostasis	Blood glucose graphs: identify insulin peak; describe mechanism (negative feedback). Urine concentration data: explain role of ADH. Describe trend with actual values.
14	Plant Reproduction	Germination experiments. IV = temperature or light or water. DV = percentage of seeds germinated. Light is NOT needed for germination. Describe result for each condition tested.
15	Human Reproduction	Hormone cycle graphs. Identify ovulation at LH surge (day 14). Compare oestrogen and progesterone over 28 days using actual values from graph axes.
16	Genes and Proteins	Gel electrophoresis band pattern interpretation. Complementary base pairing problems. Magnification calculations from scale bars. Draw complementary DNA strand given one strand.
17	Variation	Measure continuous variation in class (e.g. hand span). Histogram bars MUST TOUCH for continuous data. Calculate mean and range. Identify type of distribution (normal = bell curve).
18	Ecology	FREQUENTLY TESTED. Quadrat sampling. Population estimate = (count in quadrat / quadrat area) x total area. Food web interpretation: effect of removing one species. Transect diagrams.
19	Human Influences	Pollution data graphs. Species diversity vs pollution level curves. Eutrophication data sequence. Always describe trends with actual data values from axes.
20	Biotechnology	Biological vs non-biological washing powder investigation. IV = enzyme type or concentration. DV = degree of stain removal (measured by colorimeter or rated visually). Fermentation rate data graphs.

P6 Worked Examples and Model Answers

Worked Calculations

Example	Step-by-Step Working	Final Answer
Magnification: cell image measures 45 mm wide. Actual cell = 0.03 mm.	Convert units: actual = 0.03 mm = 30 um. Image = 45 mm = 45000 um. M = 45000 / 30	x1500 (no unit)
% change in osmosis: initial mass = 5.0 g. Final mass = 4.2 g.	Change = 4.2 - 5.0 = -0.8 g. % change = -0.8 / 5.0 x 100	-16% (water left cell by osmosis)
Rate of CO2 production: 18 cm3 collected in 6 minutes.	Rate = 18 / 6	3.0 cm3/min
Dilution: make 0.5 mol dm-3 in 50 cm3 total from 1.0 mol dm-3 stock.	C1V1 = C2V2. 1.0 x V1 = 0.5 x 50. V1 = 25 cm3	25 cm3 stock + 25 cm3 distilled water
Population estimate: 8 organisms in 0.5 m2 quadrat. Total area = 200 m2.	Estimate = (8 / 0.5) x 200 = 16 x 200	3200 organisms
Mean of 3 repeats: 12, 14, 13 seconds.	Mean = (12 + 14 + 13) / 3 = 39 / 3	13.0 s

Model Conclusion Phrases

Experiment Type	Model Conclusion - Copy This Structure
Enzyme vs temperature	As temperature increased from 10 C to 37 C the rate of reaction increased from X to Y cm3/min. Above 37 C the rate decreased to zero because the enzyme was denatured - the active site permanently changed shape and the substrate could no longer bind.
Osmosis in potato	As the concentration of salt solution increased from 0.0 to 1.0 mol dm-3 the percentage change in mass decreased from +8% to -15%. This is because water moved OUT of potato cells by osmosis down a water potential gradient into the more concentrated solution.
Photosynthesis vs light	As the distance of the lamp from the plant decreased from 50 cm to 10 cm the rate of photosynthesis increased from 3 to 22 bubbles/min. This is because light intensity increased providing more energy for the photosynthetic reactions.
Transpiration vs temperature	The dye moved further up the xylem at 40 C (X mm) than at 4 C (Y mm) because higher temperature increased the kinetic energy of water molecules increasing the rate of evaporation from mesophyll cells (transpiration).
Yeast respiration vs glucose conc.	As glucose concentration increased from 1% to 4% the rate of CO2 production increased from A to B cm3/min. This shows yeast respires faster when more substrate is available for its respiratory enzymes.
Food test conclusion	Tube A turned brick-red with Benedicts solution after heating therefore a large amount of reducing sugar is present. Tube B remained blue therefore no reducing sugar is present.

Most Common Mistakes and Corrections

Wrong - Do NOT Write This	Correct - Write This Instead
The enzyme was killed by heat	The enzyme was denatured - the active site changed shape permanently and the substrate can no longer bind
Results increased (no numbers)	Results increased from 23 s to 41 s as concentration doubled from 0.5 to 1.0 mol dm-3
Keep everything the same (control variables)	Same volume of solution (20 cm3); same temperature (25 C maintained with water bath); same potato species
HCl is corrosive (hazard answer)	HCl is corrosive and could cause burns to skin and eyes if it comes into contact

Wrong - Do NOT Write This	Correct - Write This Instead
Use more accurate equipment (improvement)	Use an electronic water bath to maintain a constant temperature of 37 C throughout the experiment
Table with units in data cells (e.g. 10 C in cell)	Write units ONLY in column heading: Temperature / C. Data cells contain numbers only: 10
Histogram bars with gaps between them	Histogram bars MUST TOUCH because the data is continuous - gaps are only used in bar charts for discontinuous data
The potato gained or lost water (no mechanism)	Water moved into/out of potato cells by osmosis down the water potential gradient from higher to lower water potential
Circle anomaly and include it in line of best fit	Circle anomaly clearly AND draw line without passing through it AND state: the anomalous result was excluded from the line

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P6 P6 Master Summary

The Golden Rules of Paper 6

Rule	What It Means - Apply in Every Answer
1 - USE DATA	Every conclusion and evaluation answer must quote actual numbers. Results increased = ZERO. Results increased from 4 to 11 bubbles per min = MARKS.
2 - NAME CONTROLS	Never say keep everything the same. Name EACH control variable AND state how it is kept constant. Minimum 2-3 named controls in any planning answer.
3 - DENATURED NOT KILLED	Enzymes are DENATURED at extreme temperature or pH. The active site changes shape PERMANENTLY. The substrate can no longer bind. Write this every time.
4 - UNITS IN HEADINGS	Units go ONLY in column headings: Volume / cm3. NEVER write 10 cm3 in a data cell. Write 10 only.
5 - SCALE MORE THAN HALF	Graph points must spread over more than 50% of the available grid paper in both directions. Cramped scale = mark lost.
6 - HAZARD HAS TWO PARTS	Name the hazard AND state what harm it causes: HCl is corrosive and could cause skin burns = 1 mark. HCl is corrosive alone = 0 marks.
7 - SPECIFIC IMPROVEMENTS	Generic answers score zero. Repeat 3 times and calculate mean to improve reliability = marks. Be more careful = zero marks. Always name the specific action.

Complete Formula Reference

Magnification = Image size / Actual size
Actual size = Image size / Magnification
Percentage change = (Final - Initial) / Initial x 100
Rate = Quantity / Time
Mean = Sum of all values / Number of values
Population estimate = (Count in quadrat / Quadrat area) x Total area

Quick Reference - Apparatus and Units

Measurement	Correct Apparatus	Unit in Exam Answers
Length or distance	Ruler (mm scale) or metre rule	mm or cm
Volume of liquid	Measuring cylinder or graduated syringe	cm3
Mass	Top-pan balance or electronic balance	g
Temperature	Thermometer or temperature probe	C
Time	Stopwatch or timer	s or min
pH	Universal indicator paper or pH meter	pH units (1 to 14)

Measurement	Correct Apparatus	Unit in Exam Answers
Gas volume	Graduated syringe or inverted test tube in water trough	cm3
Magnification	Ruler on image + calculation	x (dimensionless - no unit written after number)
Concentration of solution	Prepared dilution series or stated	mol dm-3

Final Exam Reminders

- Read the entire experiment description before answering any questions
- Underline the command word in each sub-question: describe / explain / suggest / calculate / plan
- Check mark allocation - it tells you exactly how many separate points to make
- Time budget: 1.5 minutes per mark. 4-mark question = 6 minutes maximum
- When writing comparisons: use connecting phrases: compared to; whereas; in contrast
- Conclusion = what the data shows. Explanation = why (biological mechanism)
- For two lines on one graph: compare them at the SAME x-axis value using actual numbers
- Meniscus: always read the volume from the BOTTOM of the meniscus with eye level

★ ***Mark Scheme:*** *If you identify an anomalous result: circle it on the graph, exclude it from the line of best fit, state it was not included. If time allows, suggest repeating that data point to get a reliable value.*

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